CCH® Tagetik | **ACADEMY**

Advanced reporting



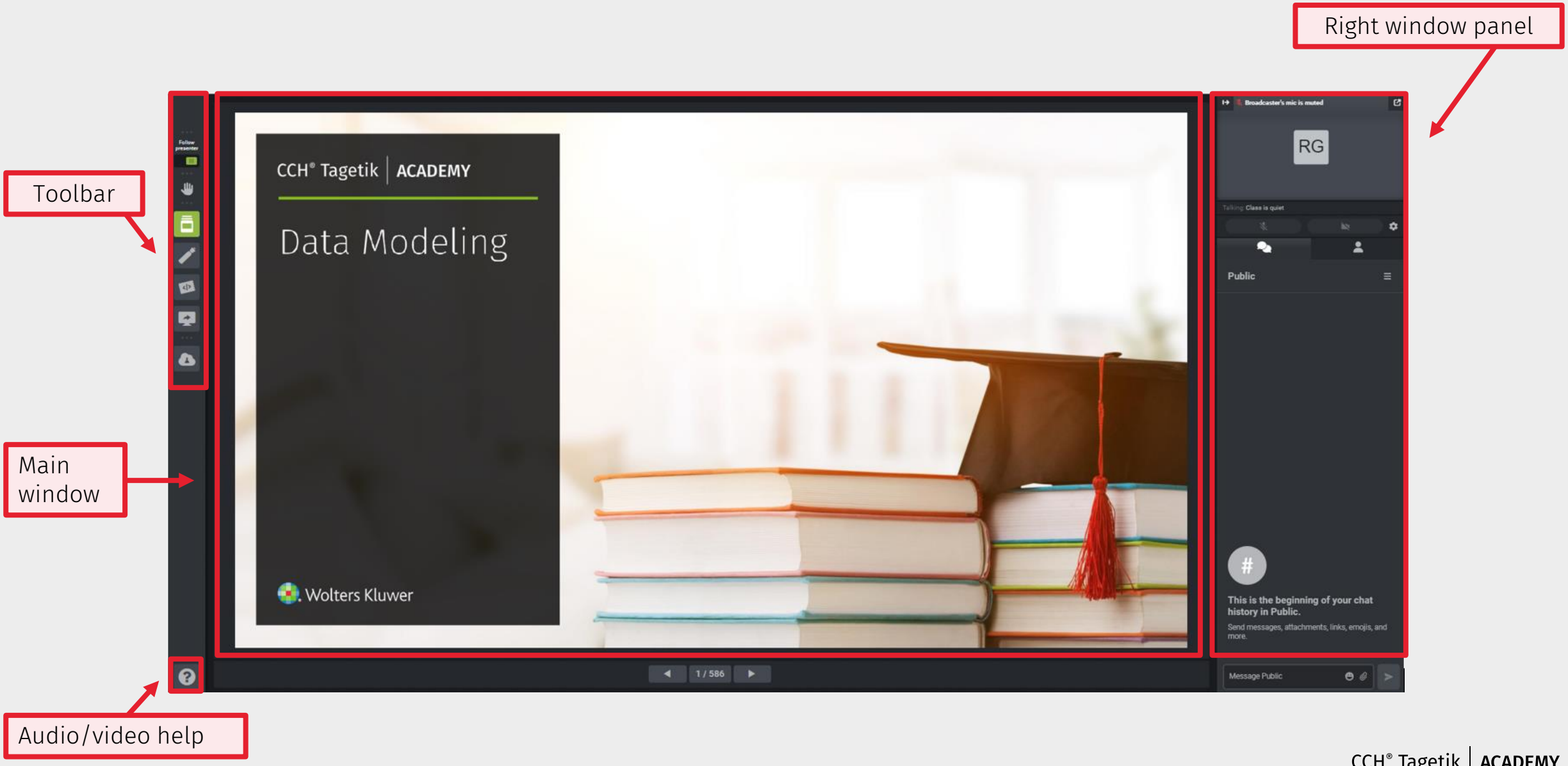
Wolters Kluwer

Before starting

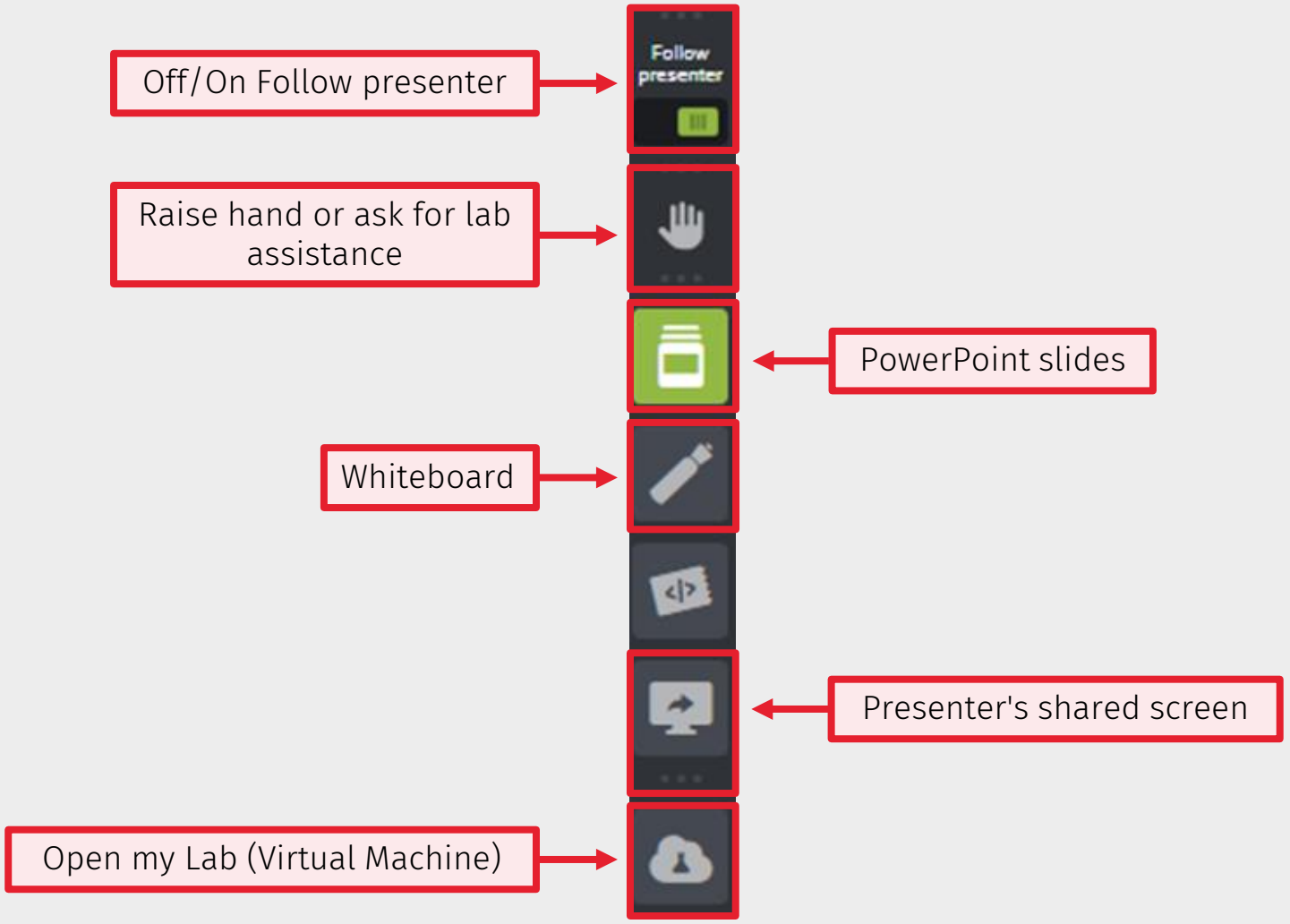
Strigo platform - “How to”



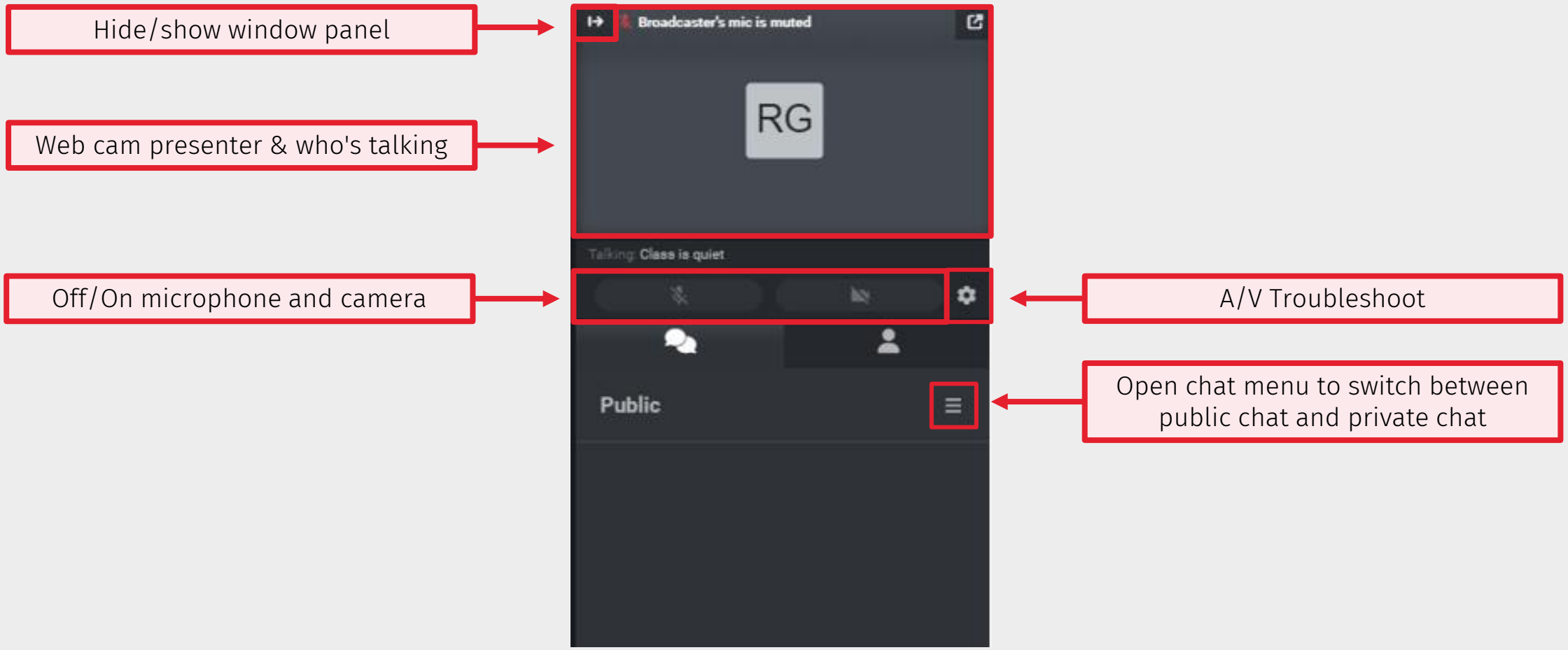
Strigo overview (for Instructor-led classes)



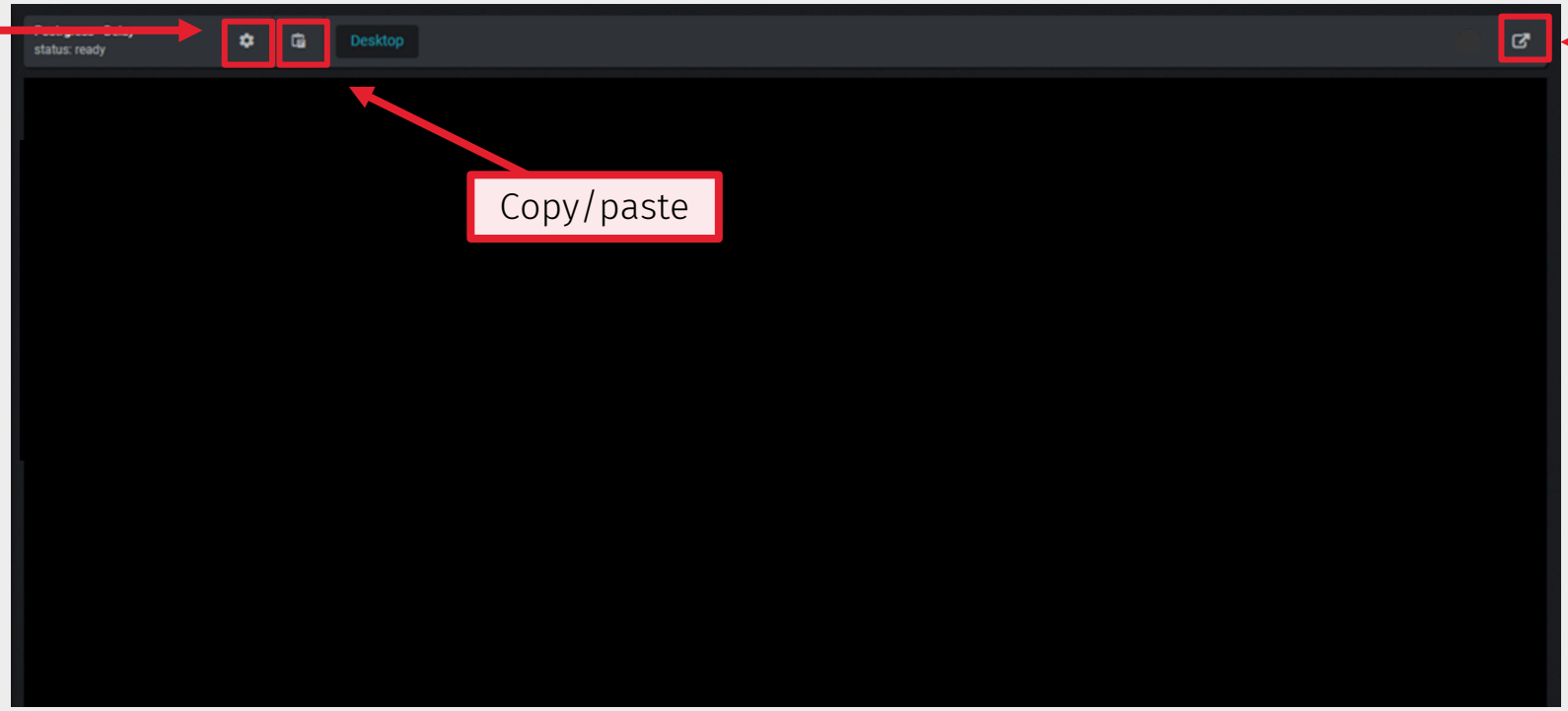
Strigo left toolbar



Strigo right window panel



Strigo Lab (Virtual Machine)



Agenda 1/2



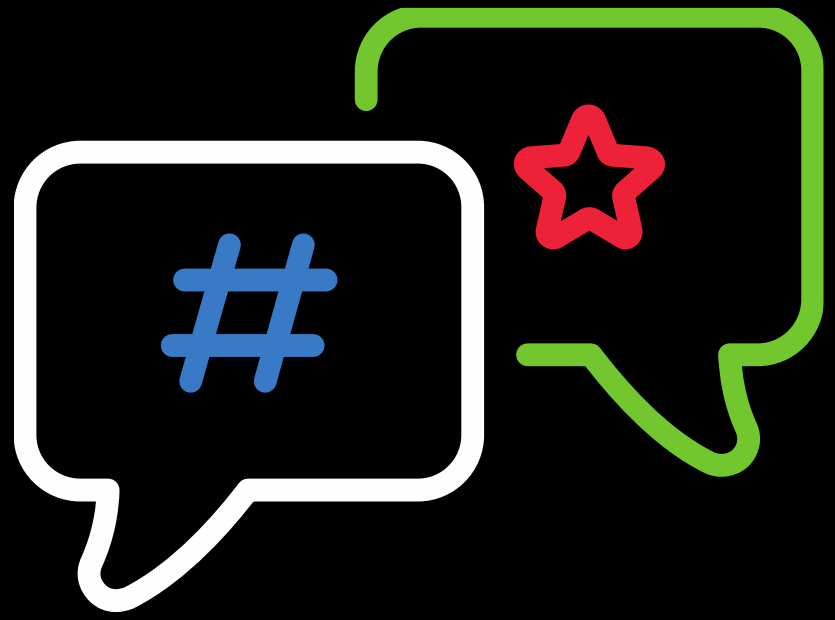
- Advanced dynamicity
- Dependent parameters
- Break back
- Reference FX rate
- Data validation query
- Best practices: optimizing forms for structure and performance
- Hide function
- Conditional editability

Agenda 2/2



- Implementation of complex data entry form
- Validation
- Data processing & Data entry forms
- Drill-through query
- Settings - Form
- Override
- FST onScenarioPeriod
- Report bursting (e-learning)

Let’s talk about: Advanced dynamicity



Dynamic forms recap

- Useful in forms including dimensions that require frequent maintenance
- Applicable on nodes of Row/Column/Tab filters
- Nodes will be deployed at runtime depending on Dynamicity type
- Tagetik style is the only way to format dynamic matrices
- Expanding methods

Dynamicity Properties

Dynamicity types

- Lowest level elements
- Children
- Advanced Dynamics

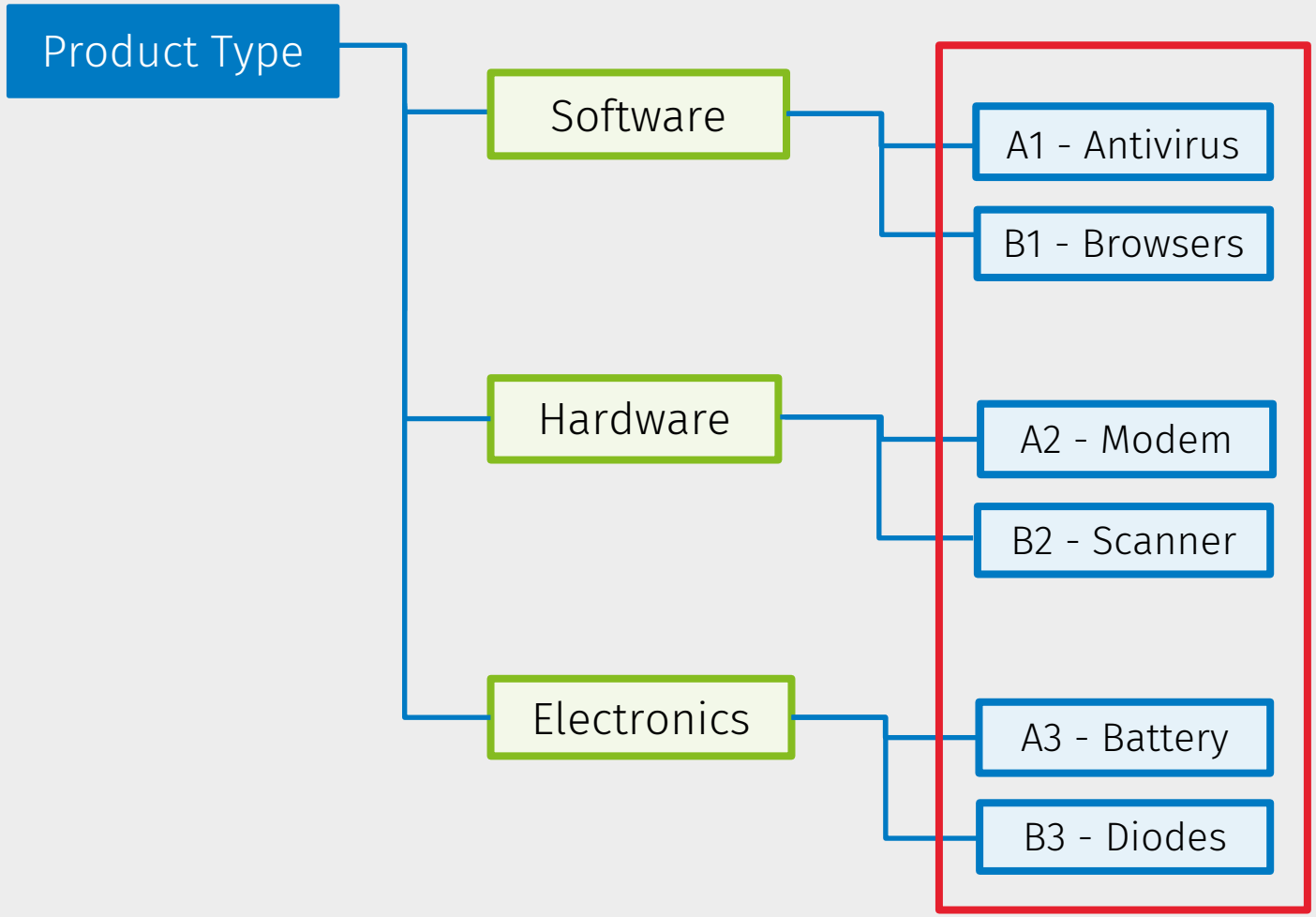
Script

- Only available for Advanced Dynamics

Pruning

- None
- Not valued
- Zero amounts

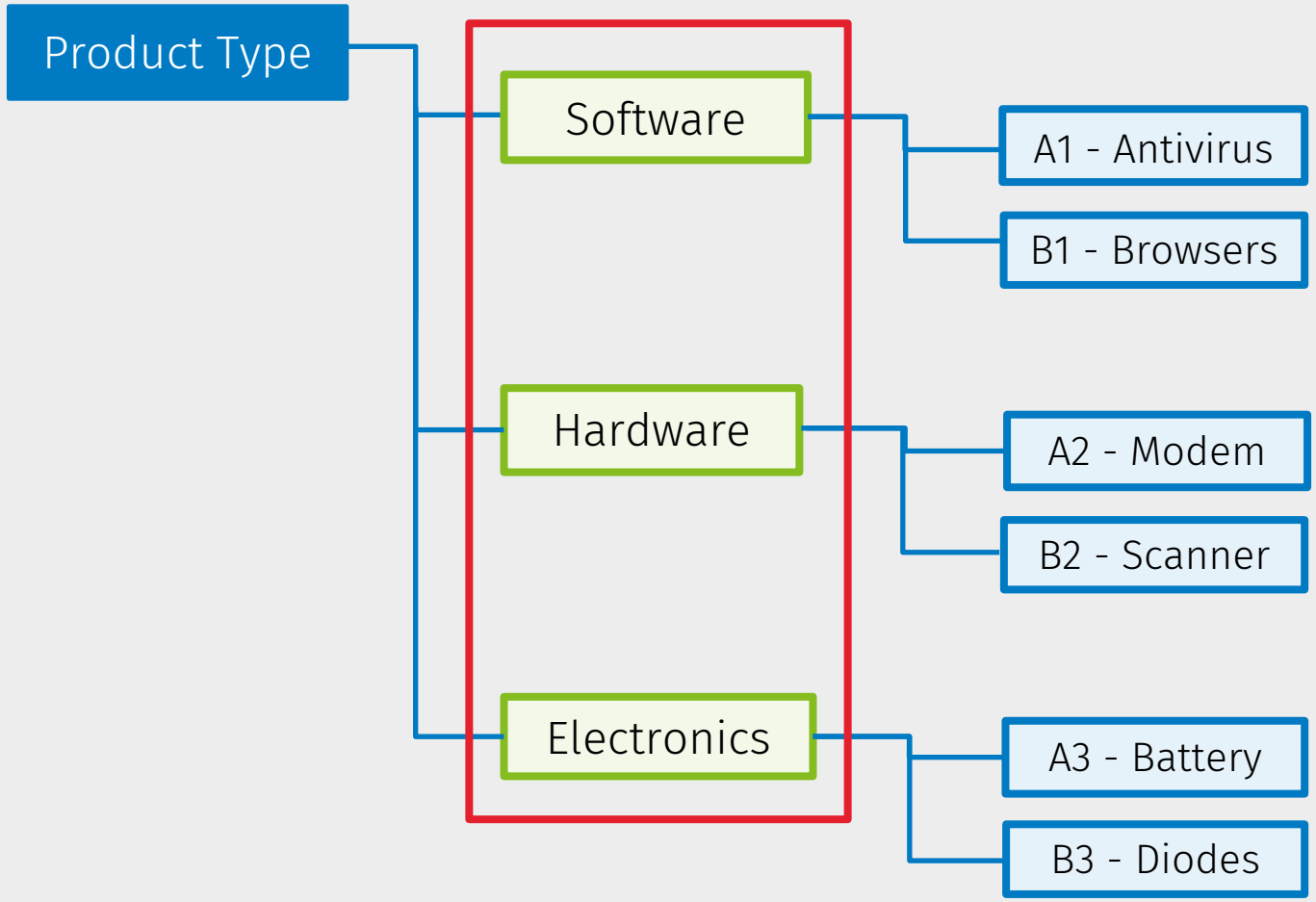
Selection results (Lowest level elements)



LOWEST LEVEL

Predefined

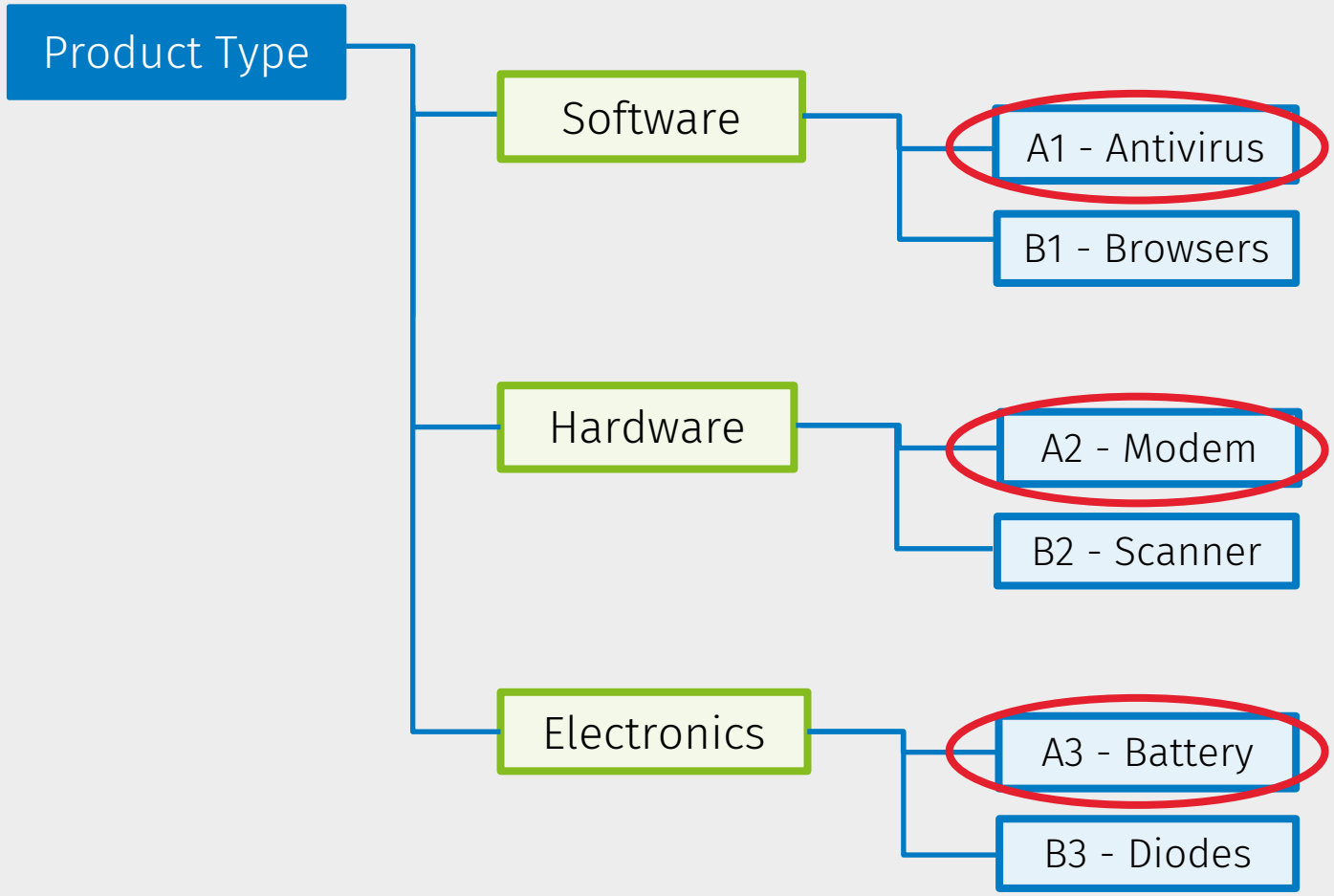
Selection results (Children)



CHILDREN

Predefined

Selection results (Advanced selection)



Advanced selection

! TGKML script required

```
TGKML 1.0

// Elements whose code starts with A

result:= this.lowest.select……;
```

Advanced Dynamicity Scripts

- Used to generate custom selection
  - E.g. “all Product elements where code starts with letter A”

- Scripting language TGKML required
  - No limit on number of assignments on local variables
  - Cell field formula allowed in scripts
  - Result of the script is a set of Dimension elements

TGKML Syntax

```
TGKML 1.0

// Comments

A:= this.lowest;


result:= A;
```

Every script starts with keyword TGKML 1.0

Add comments using the “//” notation

Keyword “this” is the node on which selection is defined

Every statement must end with “;”

Every script ends with assignment of a dimensionality to result:=.....

⚠ TGKML language is case sensitive

TGKML Functions

DIMENSIONAL FUNCTIONS

```
A:= $Account;
B:=$Account.code;
```

NON-DIMENSIONAL FUNCTIONS

```
A:= Form.code;
B:= User.code;
```

POSITION FUNCTIONS

```
A:= row("MATRIX 1",5).$entity
```

TREE NAVIGATION

```
A:= this.lowest;
B:=this.children;
```

EXCEL FUNCTIONS

String manipulation, Logical, Aggregation, Arithmetic

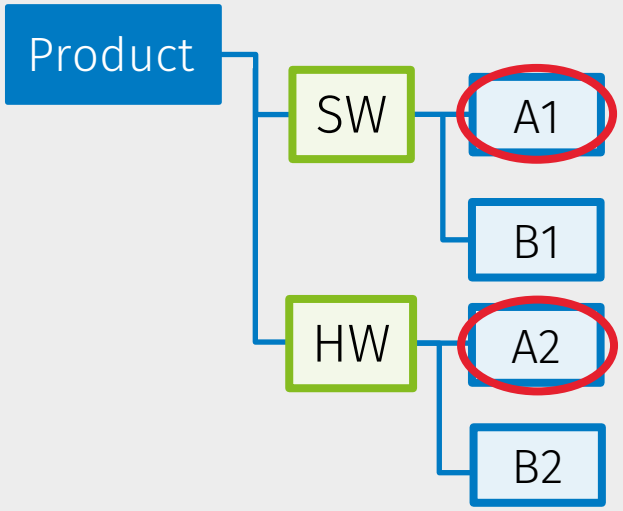
FUNCTIONS ON SETS

OR, AND, CONTAINS, etc…

TGKML “select” FUNCTION

Select has two types of Syntax

- **select(**“Expression“, “Operator“,“Value“**)**
- **select(**“Condition“**)**




```
TGKML 1.0

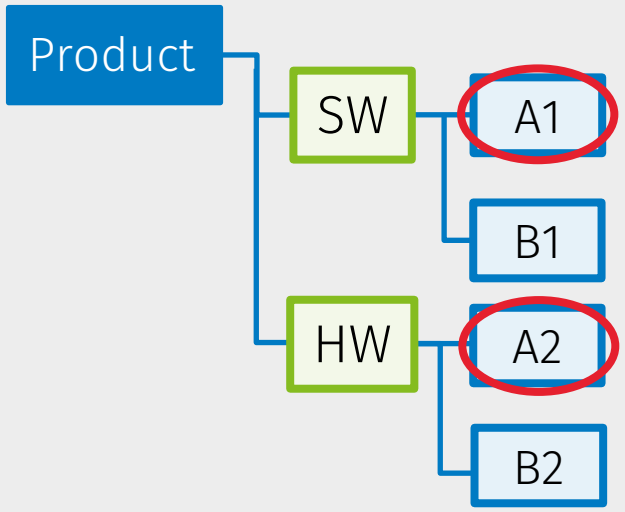
// Elements whose code starts with A
// starting from Product


result:=
this.lowest.select(“LEFT($Cust_Dim2.code,1)”,”=”,”A”);
```

TGKML “getLeaf” FUNCTION

Syntax:

```
getLeaf("code")
```




```
TGKML 1.0

// Return product with code A1


result:= this.getLeaf(“A1”);
```

TGKML if....then.....else.....end

Syntax:

```
if (condition) then Instruction1 else Instruction 2 end;
```

```
TGKML 1.0

// Result depends on condition

a:= this.lowest;

b:= this.getleaf("A1");

result:= if($Entity.code == "SLS") then a else b end;
```

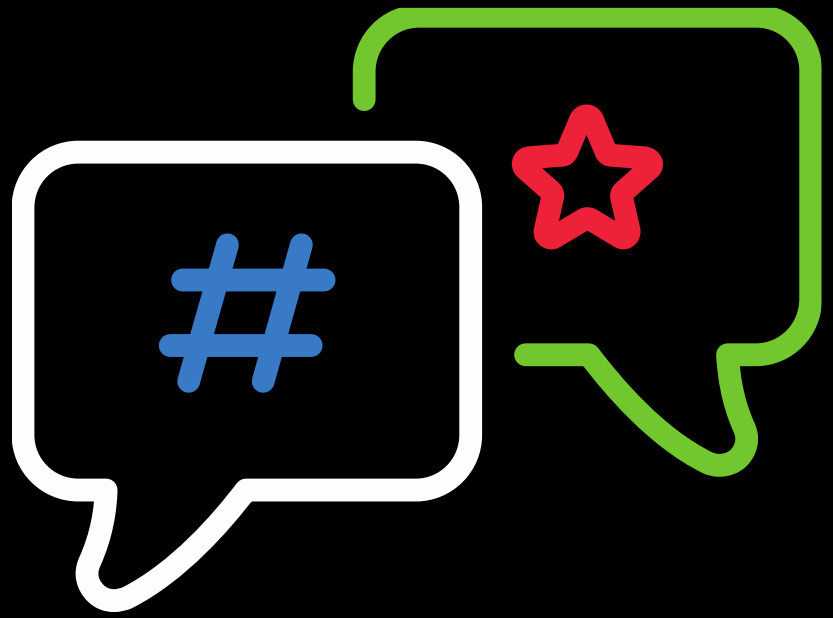
Now it’s your turn

Exercise # 1

Advanced Dynamicity



Let’s talk about: Dependent parameters



Custom Parameters & Dependent Parameters

Customize list options

- Manually
- Dynamic

Dependent parameters

- Scripting
- Manual order

Non-dimensional types

- Flag
- Date
- Text
- Number
- Dictionary

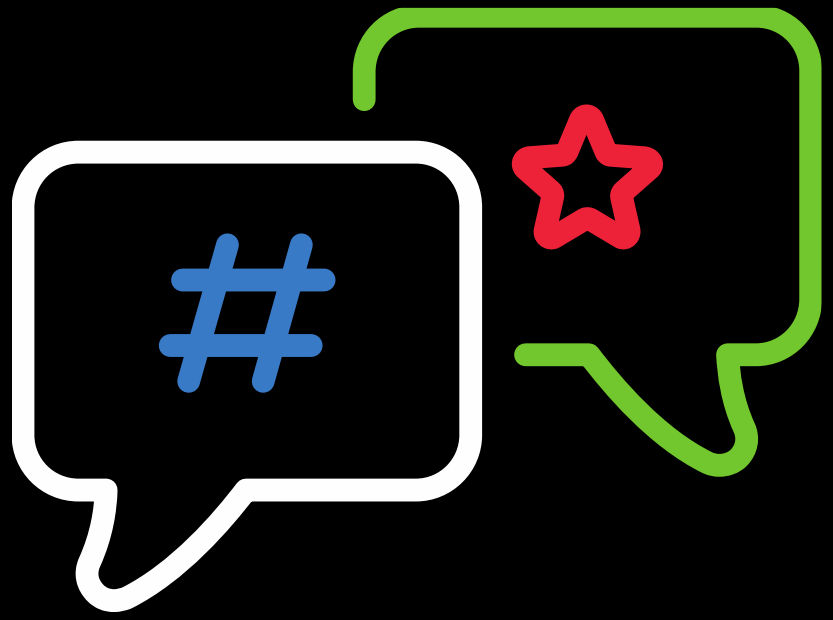
Now it’s your turn

Exercise # 2

Dependent parameters



Let’s talk about: Break back



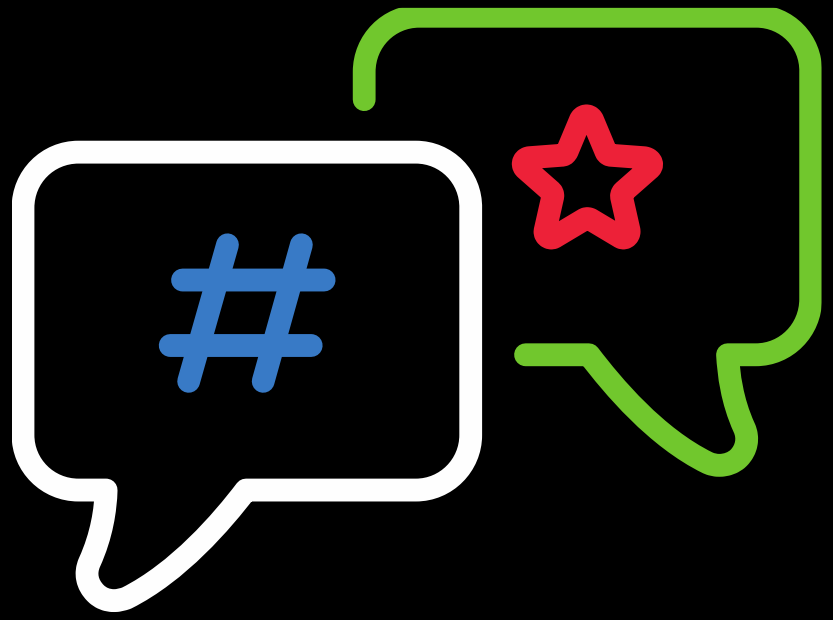
Break-Back

Used when Top Down approach is necessary to draft Budget

Allows you to enter data at hierarchy node level in Data Entry forms

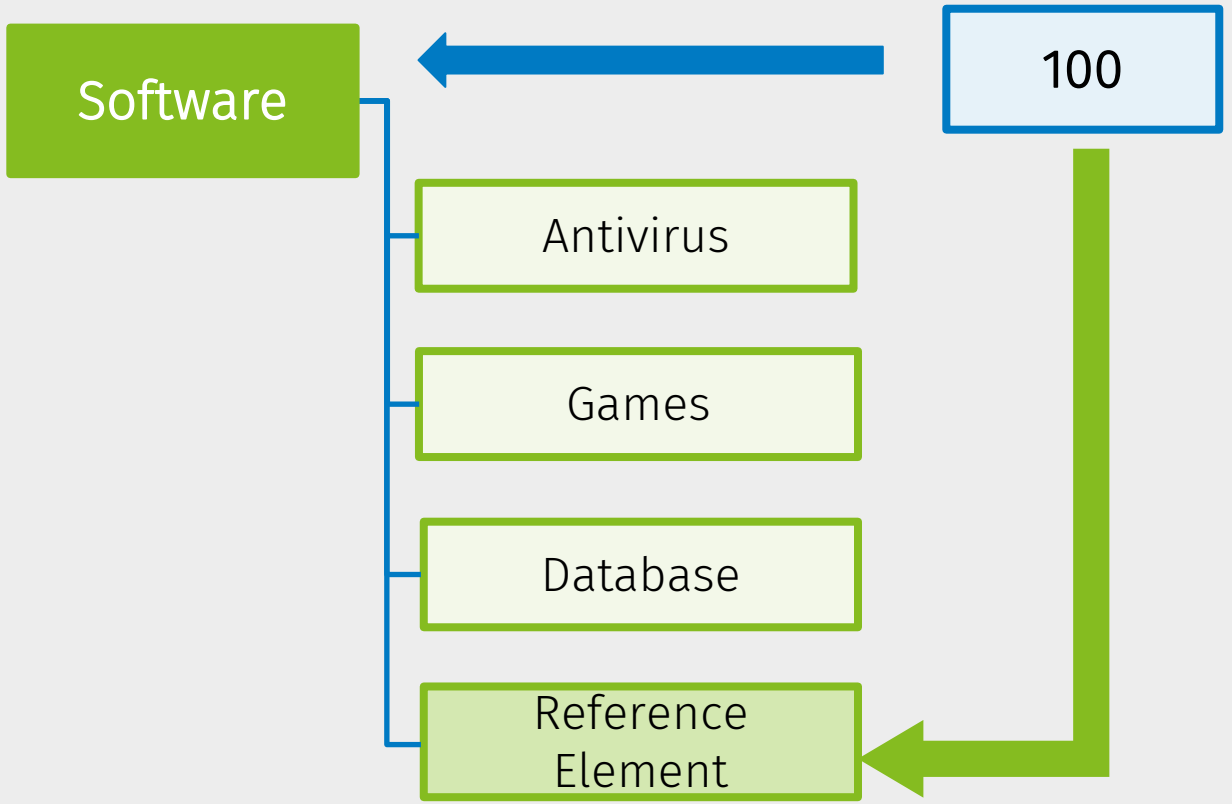
Amount will be distributed on lowest level elements

Let’s talk about: Break back on AIH

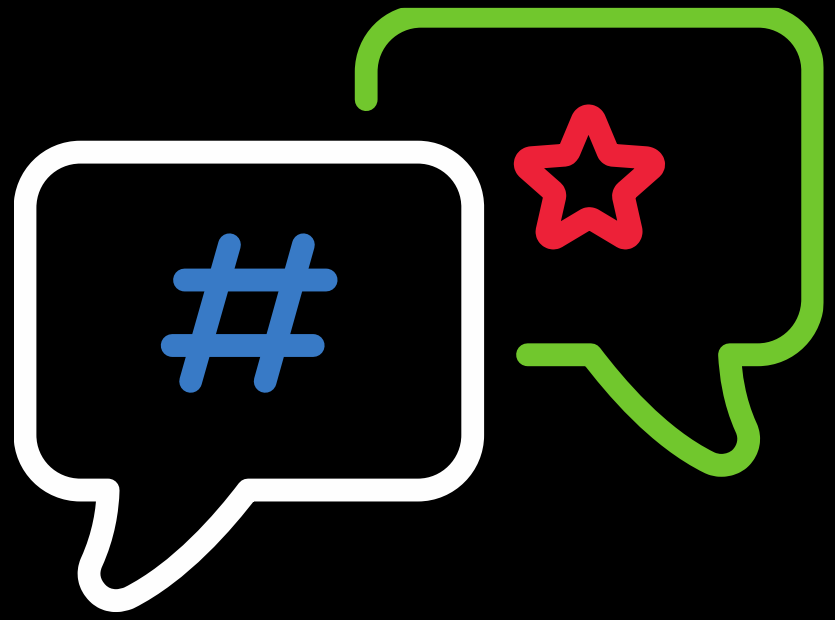


Break-back on AW multidimensional matrices: save on reference element

- Data entered on a node saved on a reference element
- Reference element must be defined in each node
- Always enabled for **Custom Dimensions, Analytical dimensions, Entities, Accounts**



Let’s talk about: Break back settings on FW



Break-Back on FW

Can be enabled for:

- None (Break Back Disabled)
- Custom Dimensions
- Custom dimensions, Entities and accounts

Break-Back setting: Enable/Disabled

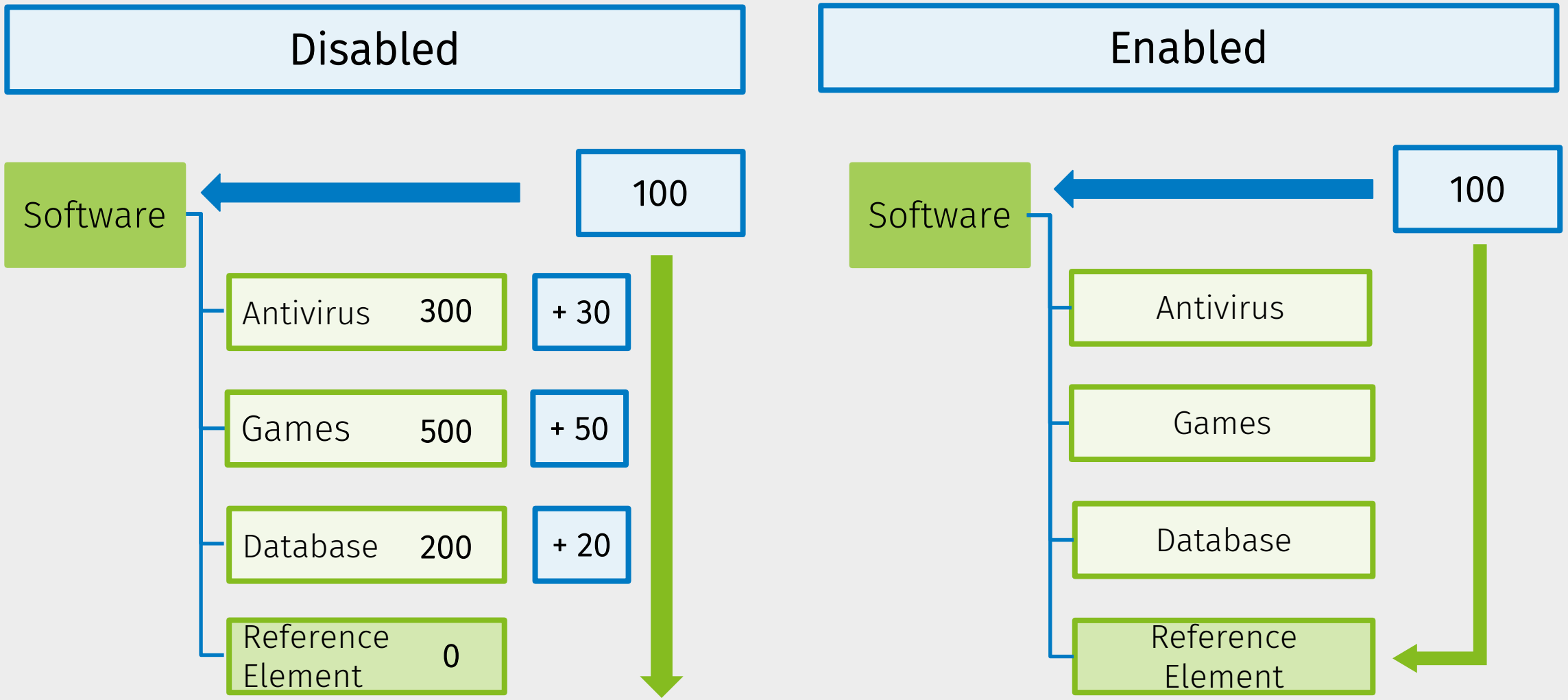
Disabled

	January
Software	
Hardware	
Electronics	

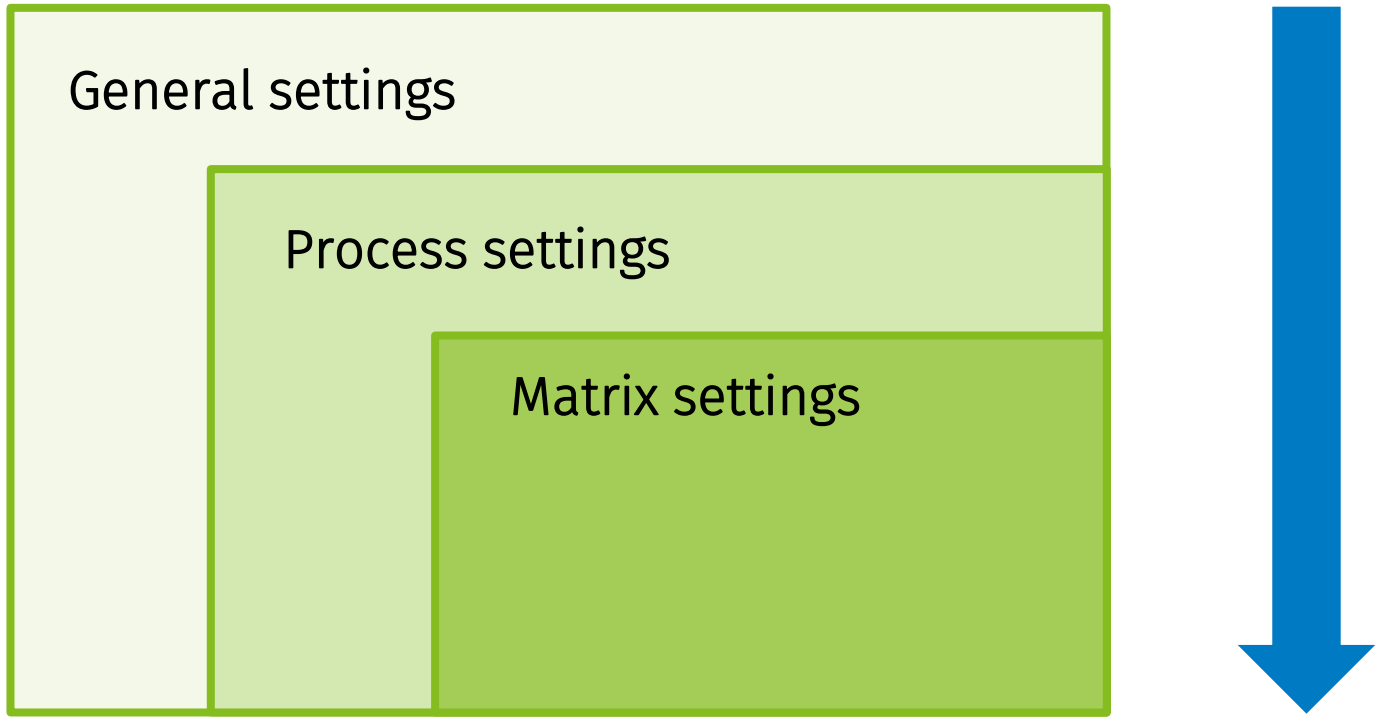
Enabled

	January
Software	100
Hardware	30
Electronics	250

Break-back setting: save on reference element



Break-back settings level and override



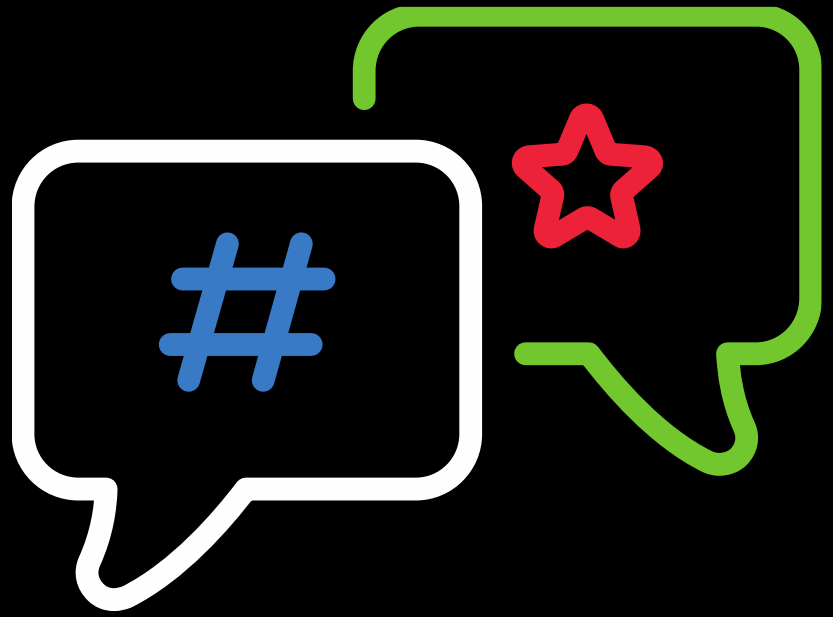
Now it’s your turn

Exercise # 3

Break back



Let’s talk about: Reference FX rate



Reference FX rate

It is often necessary to compare data of a given scenario/period by converting them to different FX rates

Necessary to specify Scenario/Period and FX rate type

Supported for FW matrices

Reference FX Rate : use case

FX rate impacts on figures, requires a proper analysis in case of Currencies with big fluctuations

	Revenue Peso	Peso/USD FX rate	Revenue USD
Current year	150	10	**15**
Previous year	100	5	**20**



Create a form that separates operational component from FX rate impact

Reference FX Rate : use case

	Previous year	Current year with previous FX rate	Current year	Variance	Operational variance	FX Impact
Amount (USD)	20	**30**	15	15-20= **-5**	30-20= **10**	-5-10 **= -15**



Use of FX rate different from Running Scenario/Period

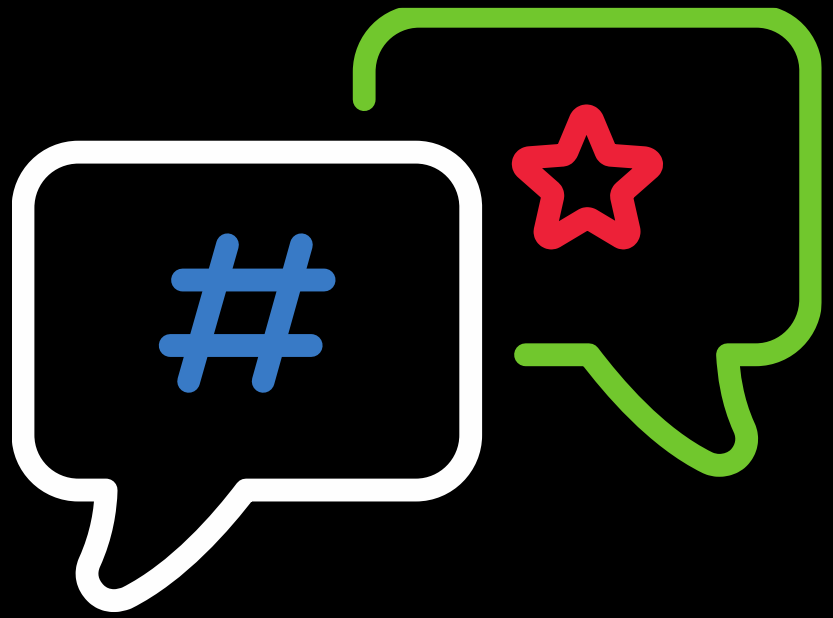
Now it’s your turn

Exercise # 4

Reference FX rates



Let’s talk about: Data validation query



Data Validation Query

Used in Data Entry forms

- It is necessary to **limit the choice of values** in matrix editable cells
- Choice is made using **combo boxes**
- Combo box values are retrieved from a **Data validation query**
- Typically used to fill cells that collect **Text or Dates**

Data Validation Query features

- Defined in the form
- Result not displayed in enabled combo boxes or in Excel sheets

Data Validation Query: Typical implementations

Query can retrieve data from:

- Dictionary
- Tagetik tables (Example: Dimensions)

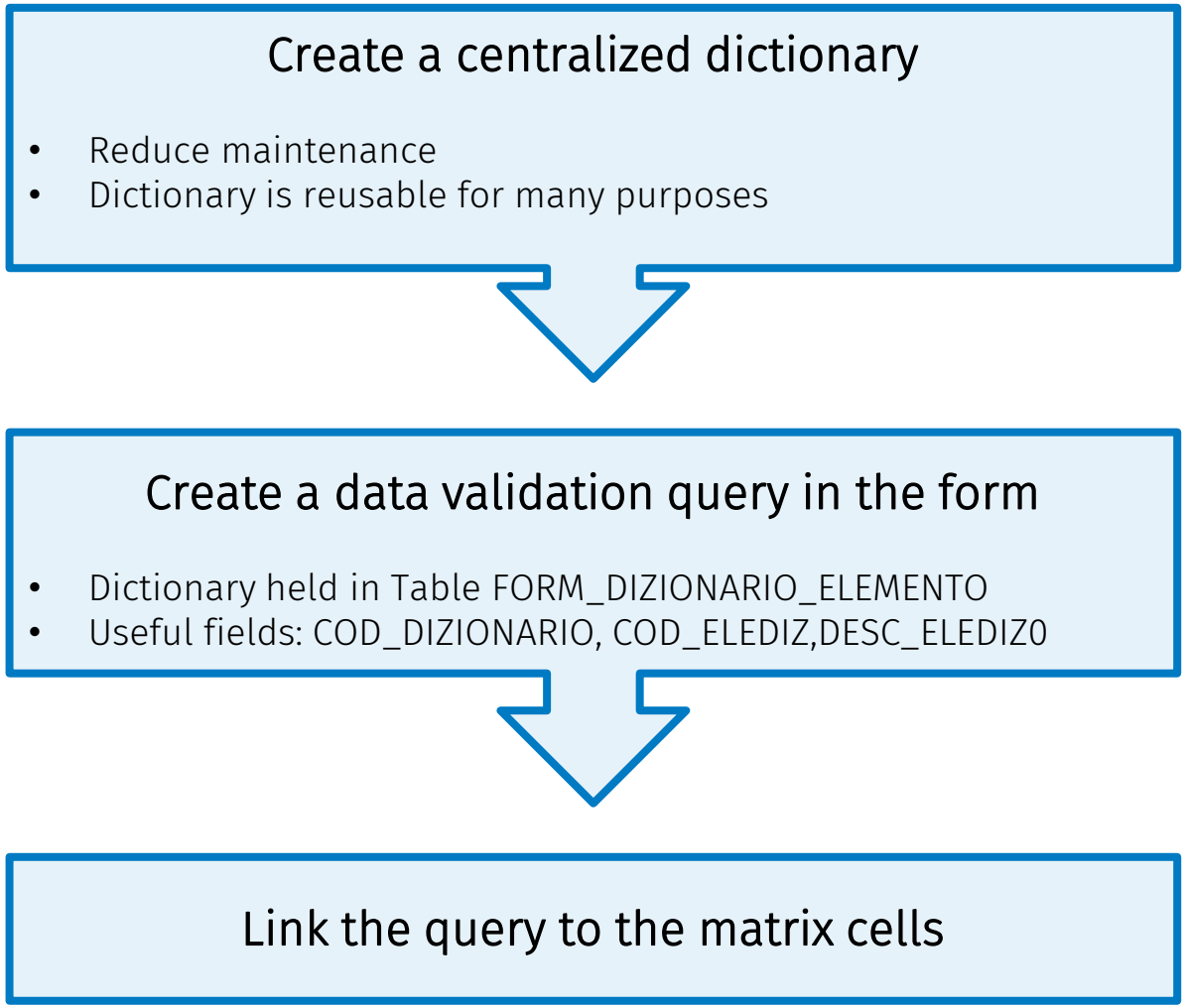
Query can generate a fixed list of elements:

- Select “Yes” union all select “No”

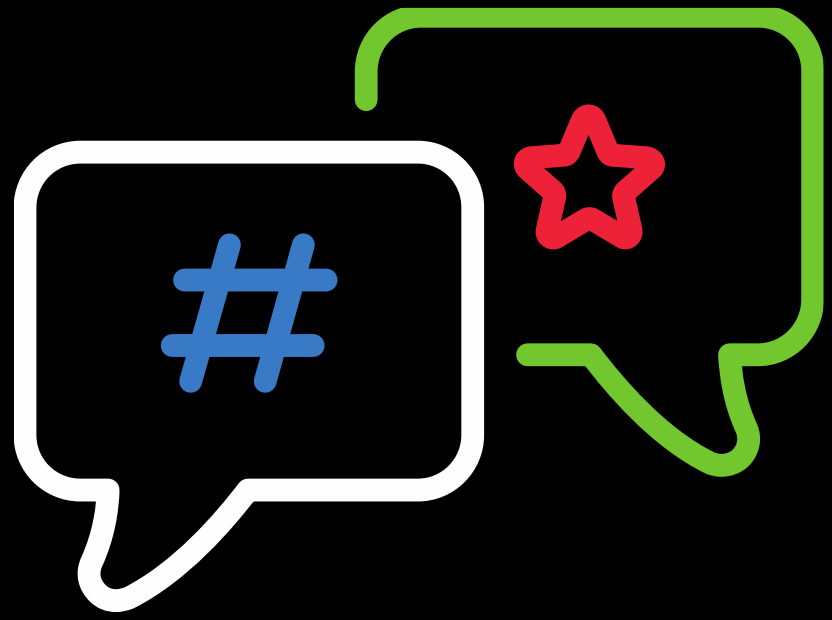
Examples list elements:

- 0/1
- Yes/No

Data Validation Query from dictionary– How to configure it



Let’s talk about: Data collections of Text fields



Data collection of text fields

Financial Workspace

- **Data storage:** DATI_NOTE
- **Data collection of multiple text fields:** store each field in a specific category

Analytic Information Hub: multidimensional matrixes

- **Data storage:** dataset measure fields
- **Data update mode:** Update existing record

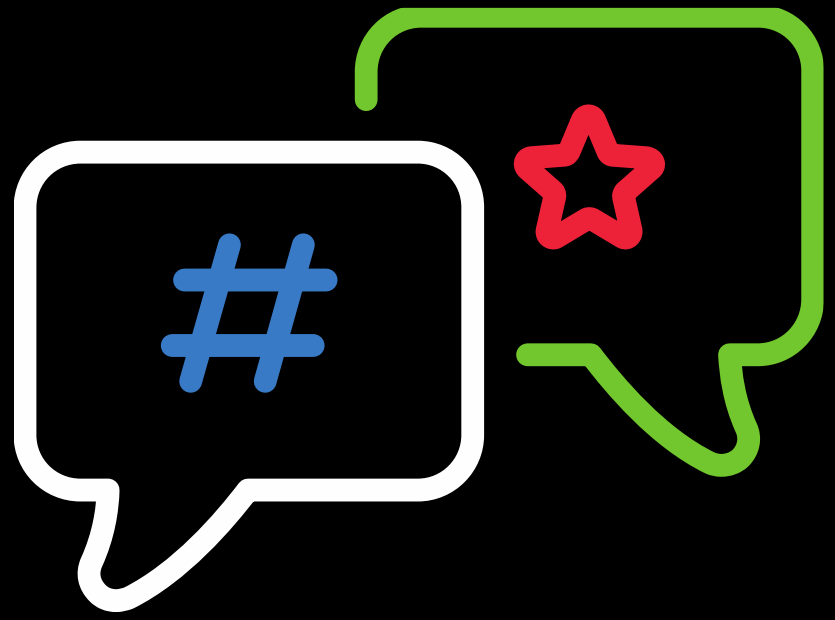
Now it’s your turn

Exercise # 5

Data validation query

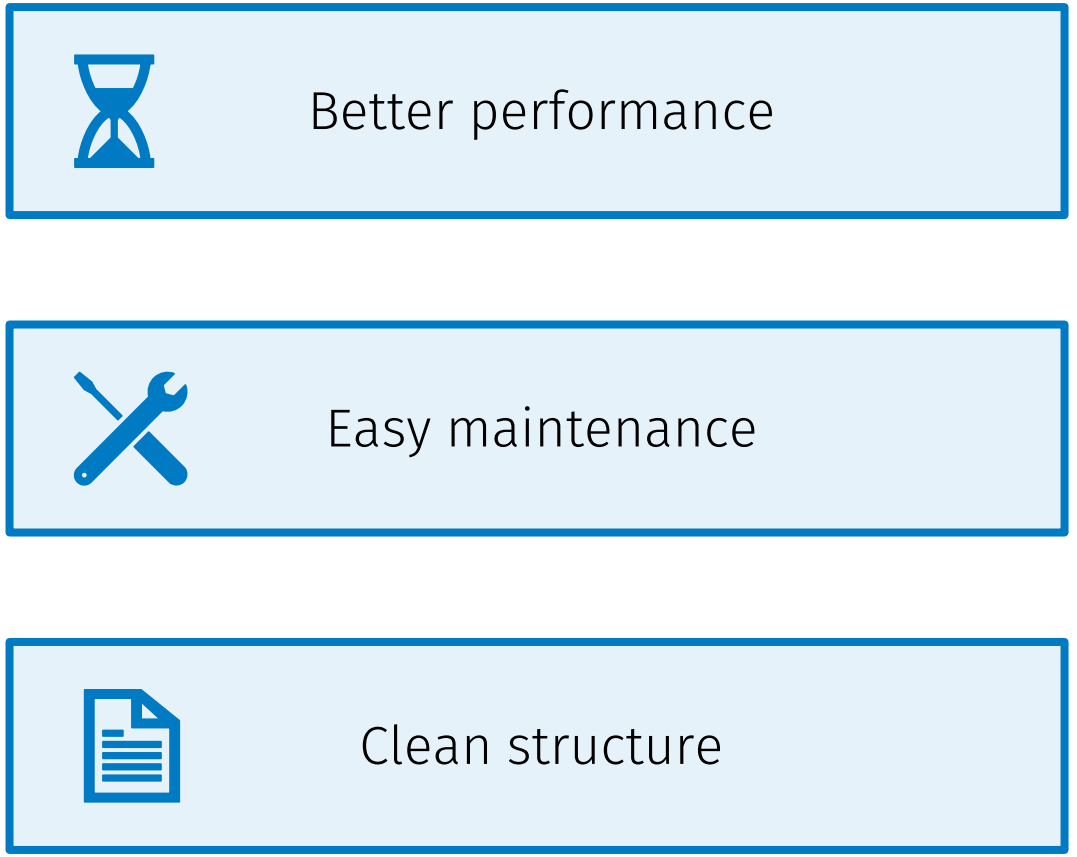


Let’s talk about: Form Optimization

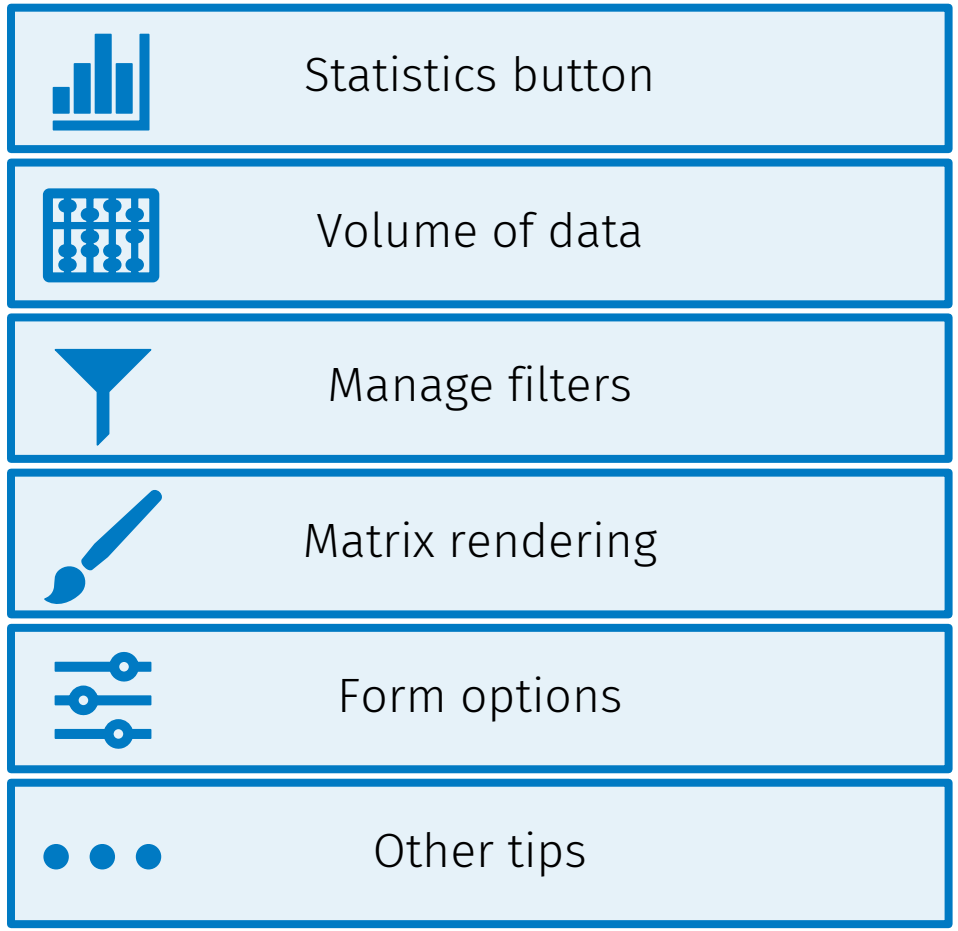


Form optimization

Optimized forms are important to have:



Main factors of performance



How to analyse performance? Use Statistics button!

Activity	Calls	Inclusive time	Inclusive %
Run Form	1	5.15	100.0%
Client activity before run	1	0.00	0.1%
VBA Macro TGK_BeforeRun	1	0.00	0.1%
Run	1	5.09	98.9%
Retrieving remote tracer	1	0.03	0.6%
Server - Total execution time	1	2.88	55.9%
Network overhead, serialiation/deser, queue time	1	0.17	3.3%
Unpacking elaboration result	1	0.04	0.8%
Workbook downloading and opening	1	0.84	16.3%
Style sheet inclusion	1	0.52	10.2%
Rendering report	1	0.53	10.4%
Client activity after run	1	0.05	1.0%
VBA Macro TGK_AfterRun	1	0.00	0.0%
Excel document activation	1	0.02	0.3%

Time Spent

Running Macros

Querying Data

Transferring Data

Rendering Report and Running Transformations

Running Macros

Volume of Data

AIH

- Data partitioning
- Efficient dataset design

FW

Archive data:

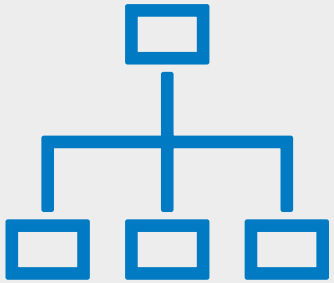
- Move data to H_DATI_*Tables

Data maintenance:

- Compress
- Delete scenarios
- Do not save incrementally

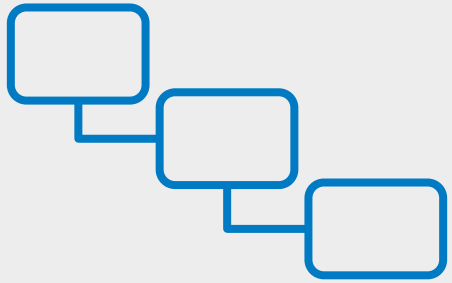
Manage Filters

- Big impact on performance
- Every filter result affects the server query with a “where” clause



Filter Level

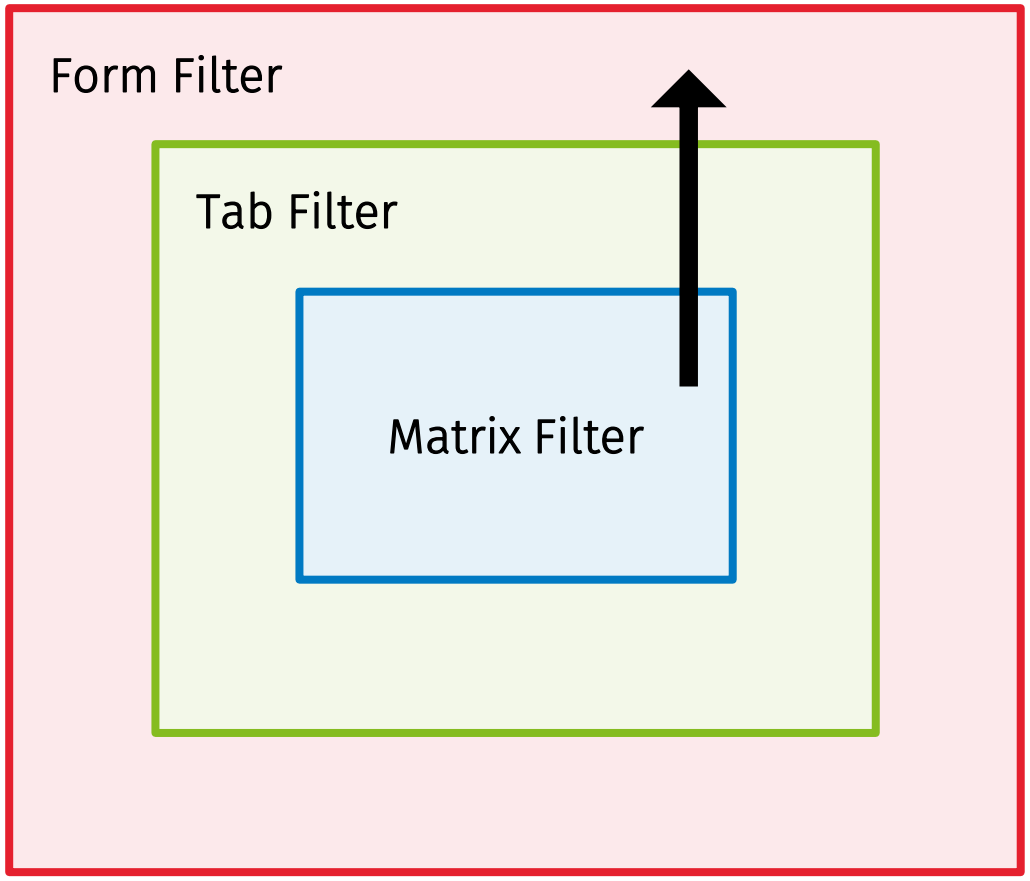
Move to the Highest Level Possible



Nested Dimensions

Avoid Unless Strictly Necessary!

Filter Level - Recommendation

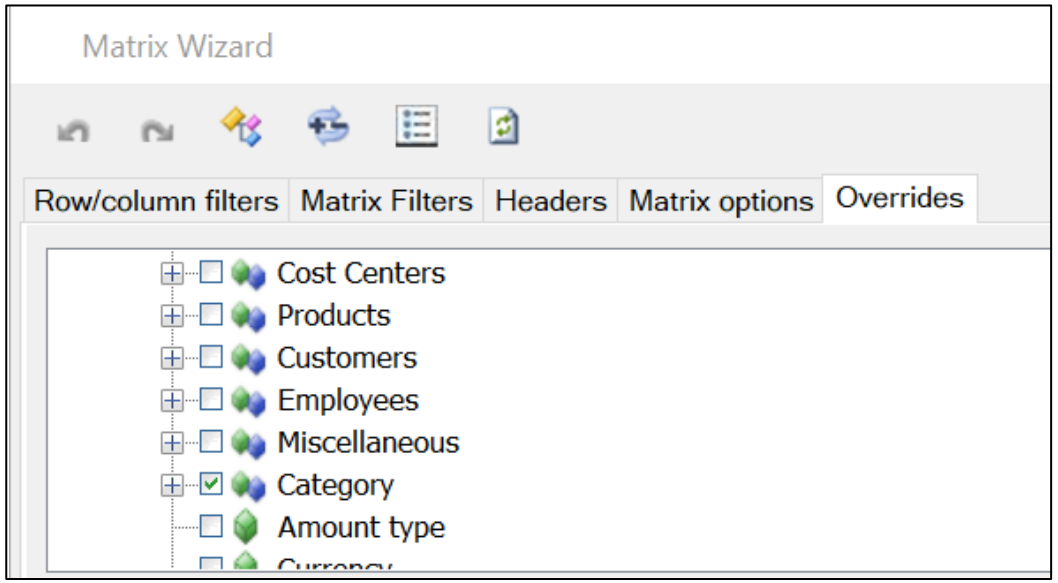


The more we can move the Filter towards the “top” the better performance!

Matrix Overrides

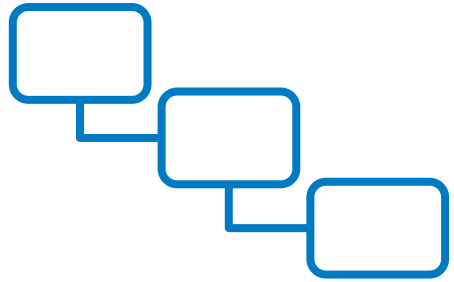
What if I need to set a filter to all matrices except one?

Use a global filter and then set an “override” on the exception matrix

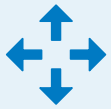


Nested Filters – Recommendation

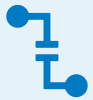
The more dimensions are nested the slower the report will be!



Alternative to nesting



Move the filter out of Row/Column and into Tab/Form



Split the matrix into 2 or more and use named ranges

If nesting cannot be avoided...



Place Static and fewest elements at the top



Use Pruning (N.V) and to zero

Matrix rendering



Styles

Only Apply if Matrix is Dynamic



VBA Scripts

Bad Scripts can have negative effects



Complex header

Consider Using Excel Formulas Instead



Report template

Do not store template in “Page” view

Form Options

Data processing type

- Synchronous (time out error)
- Asynchronous

Choice of reporting engine

- Standard
- In memory

Concurrency level

- Allows you to set the maximum number of activities which can be run concurrently

Database hints

- Hints are the mechanism which the database engine uses to generate a query plan to retrieve the data

Data processing type

Setting applicable to:

- Run form
- Save form
- Run data processing

If data processing takes....	Then set...
A long time to run or save	Asynchronous mode
A short time to run and save (less than 1 minute)	Synchronous mode

Choice of Reporting engine

Standard



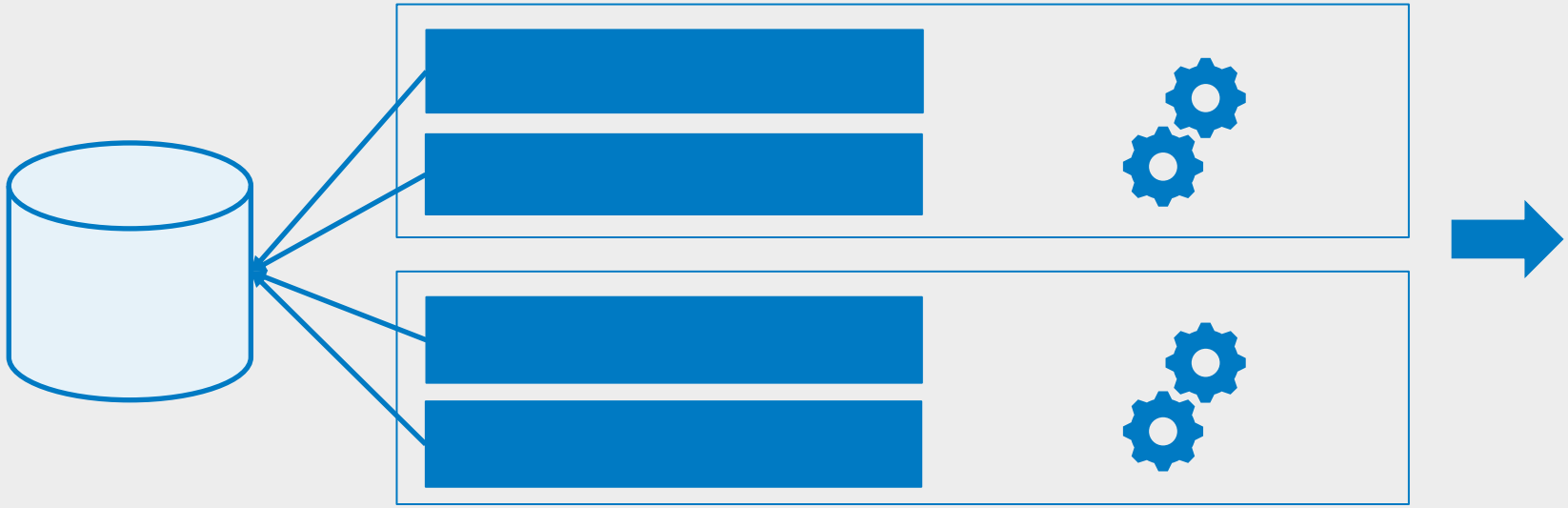
- Complex SQL queries are run according to the desired filters
- Data is transformed and refined directly as part of the SQL Query

In-Memory



- Simple SQL Queries are run to retrieve the data “in bulk” according to the filters
- Data is transformed and refined directly in memory

In Memory Reporting – How does it work?



	2017 Actuals	2018 Budget
	December	December
Sales Revenue	113,818,537	897,036,274
Discounts & Promotions		
TOTAL NET SALES	**113,818,537**	**897,036,274**
Cost of Goods Sold	1,070,128	216,092
GROSS MARGIN	**112,748,408**	**896,820,183**
Wages & Salary	7,015,120	14,299,663
Payroll Taxes	455,063	963,502
EE Benefits	426,842	570,964
TOTAL EE EXPENSES	**7,897,025**	**15,834,130**

Database

Data is retrieved from the Database via queries

For each matrix a number of connections is opened

Data Blocks

Data is grouped in data blocks of fixed size to be processed in-memory

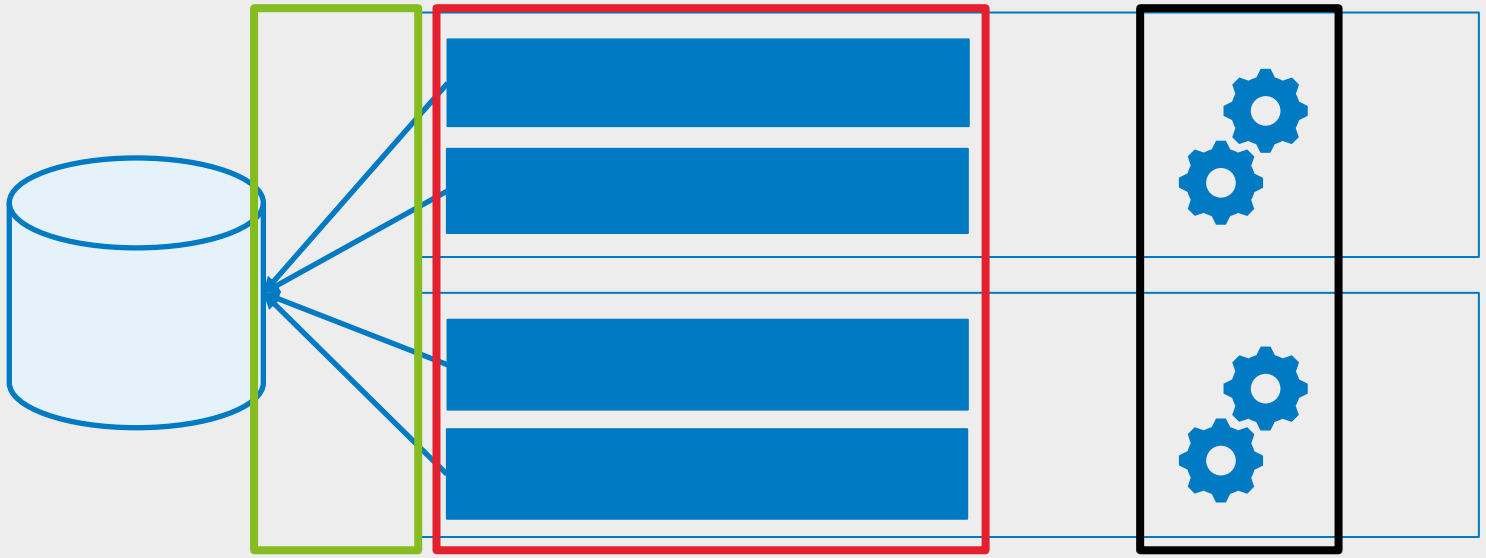
Data Processing

Data is processed sequentially or concurrently to create the data for the output matrix

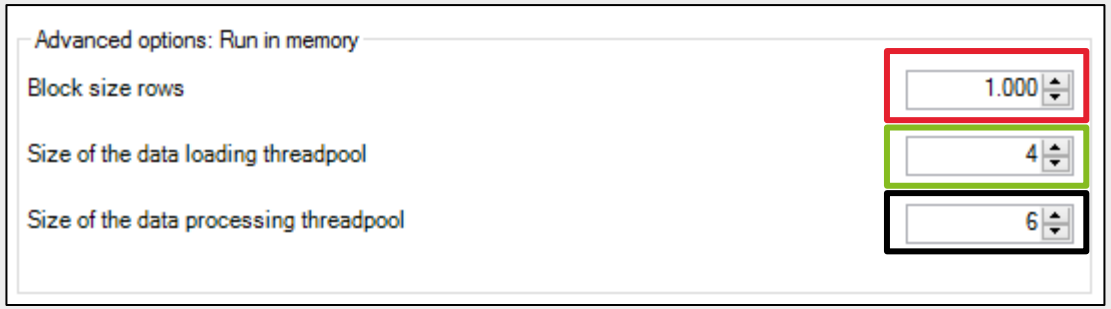
Output Matrix

Output Matrix is rendered as normal

In Memory Reporting – How can it be tweaked?



	2017 Actuals	2018 Budget
	December	December
Sales Revenue	113,818,537	897,036,274
Discounts & Promotions		
TOTAL NET SALES	**113,818,537**	**897,036,274**
Cost of Goods Sold	1,070,128	216,092
GROSS MARGIN	**112,748,408**	**896,820,183**
Wages & Salary	7,015,120	14,299,663
Payroll Taxes	455,063	963,502
EE Benefits	426,842	570,964
TOTAL EE EXPENSES	**7,897,025**	**15,834,130**



Size of the data blocks containing the Data

Maximum Number of DB Connections per Matrix

Number of Data Blocks Processed Concurrently

Choice of reporting engine

If report has a lot of	Then use…
• Filters • Matrices • Tab • Nested elements	Standard engine
Data transformations as: • On-the fly conversion • Segment • Time-dependent hierarchies	In memory engine

But… There is not a definitive answer! Use trial and error!

Concurrency level

Allows you to set the maximum number of activities which can be run concurrently

Element to run concurrently

- Matrices, Templates, Queries
- Template + Matrices suggested choice

Concurrency level

- High Concurrency will increase performance of single form but could overload the server!
  - General Rule: Concurrency should not be > 4
  - Has an impact on running in parallel elements (Matrices) with the same size (number of cells)

Database hints

Mechanism by which the database engine uses to generate a query plan to retrieve the data

Settings available

- **Enable:** Database Hints Enabled (Default Setting)
- **Enable Calculation on Data:** Hints are run based on the content of the data tables
- **Disable:** Database Hints are disabled
- **Automatic:** Setting is decided by the system at run-time

Guidelines

- Sometimes this will improve performance, but this is not a given → Trial and Error Approach
- The recommended option is to leave Hints enabled

Other tips

Advanced selection

- Advanced selection for sorting is time consuming

Hierarchy and filters

- Select filters from hierarchy instead from flat list
- Avoid creating nodes with lots of elements

Superfluos settings

- Only define editability in rows or columns, not on both
- Disable Custom Functions if these are not required

Other

- line-by-line matrix is generally faster than multidimensional

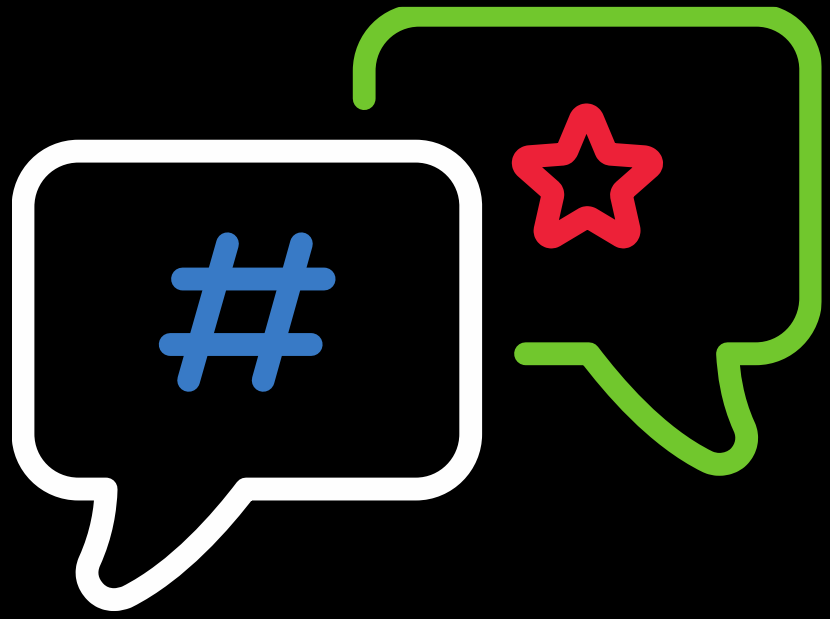
Now it’s your turn

Exercise # 6

Form optimization



Let’s talk about: Hide function



Hide function

What does it do?

- Allows you to Hide rows/columns in the presence of certain conditions when the form is run
- Rows/columns are hidden, not deleted
- Hide affects performance

Why?

- To Have a proper layout for each output sheet
- Hide matrix formulas used only for calculation and must not be displayed
- Hide rows used as fixed reference in the matrix

Hide Options

- Always
- Not value
- To zero
- Different behavior in Navigation or Data entry

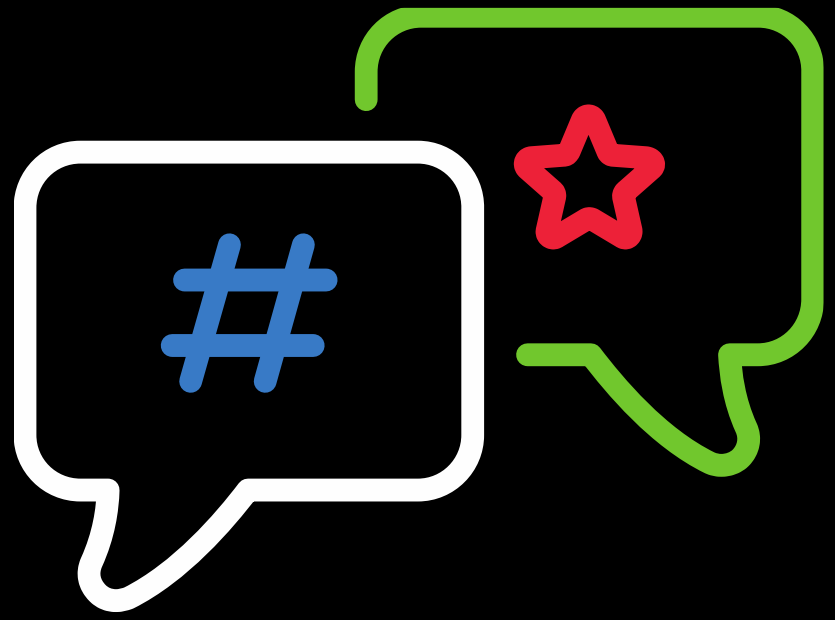
Now it’s your turn

Exercise # 7

Hide function



Let’s talk about: Conditional editability



Conditional editability

Allows to connect the actual editability to a condition based on other matrix rows /columns

Editability subordinated to one or more condition

Enabled on editable and protected rows/columns

Definition not suitable with dynamic nodes

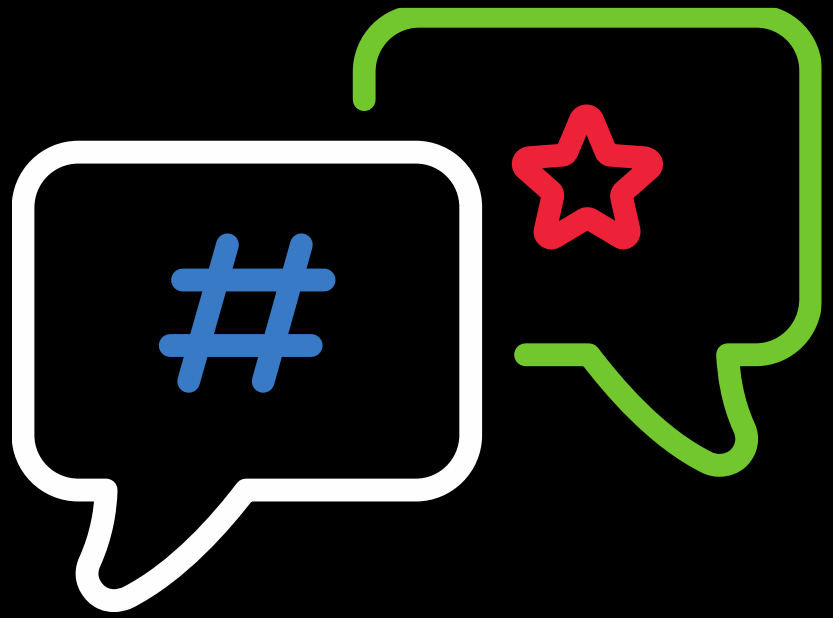
Now it’s your turn

Exercise # 8

Conditional editability



Let’s talk about: Implementing Complex Data Entry Forms



Complex Data Entry forms - Topics

What is a complex Data Entry form ?

How can data outside matrices be retrieved and linked?

How and when should in-form calculations be used

Best practices from the field

What makes a data entry form complex?

- Multiple Matrices with different dimensionalities
- Complex excel formulas
- Dynamic Matrices and Calculations

Example 1: Different formula for different column

Salary dynamic Data Entry form

	Wages and salaries	FTE	Fixed commission %
	50300	HR4000	PYL_940
Rick Deckard	91,340	1	20%
Roy Batty	135,040	1	20%
Ellen Ripley	70,400	1	35%
Charlie Bucket	55,300	1	43%
Jen Barber	45,600	1	34%
Roy Trenneman	84,000	1	45%
Total	**481,680**	**6**	**197%**

The value in Red is calculated with wrong formula!

- First and second column must display a **subtotal sum** and we need to use **`Subtotal(9, {1}:{1})`**

- Third column must display the **subtotal average** percentage and not the total. We need to use the formula **`Subtotal(1, {1}:{1})`**

Solution to the problem

The Formula will need to be different, according to the type of account being considered.

We need to:



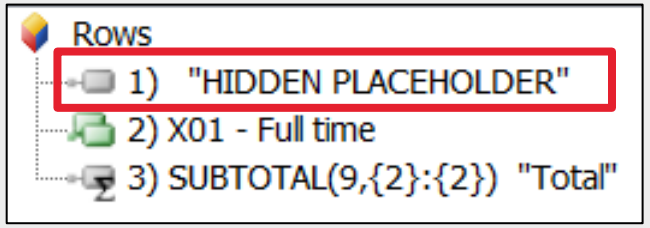
Identify the position (column) in which we are



Switch the formula based on the column in which we are

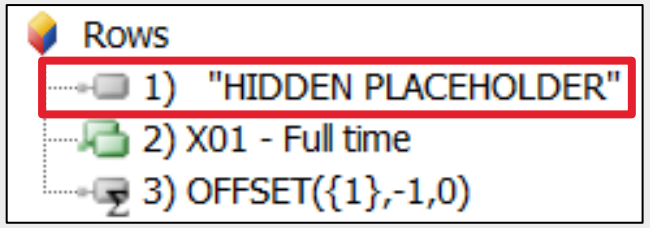
Solution A: use of hidden placeholder and OFFSET formula

1) Create a Hidden Placeholder to use as a position reference:

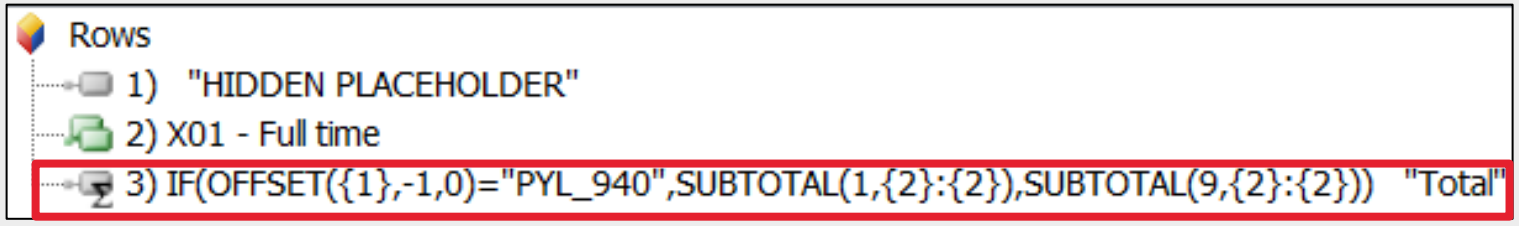


You cannot refer following nodes
because dynamicity is a dynamic node

2) Use Offset formula to retrieve the Account code: 1 row above the placeholder.



3) Use an IF Statement to switch/change the formula according to the account code.



Result

Before

	Wages and salaries	FTE	Fixed commission %
Rick Deckard	91.340	1	35,0%
Roy Batty	135.050	1	45,0%
Ellen Ripley	70.400	1	34,0%
Charlie Bucket	55.300	1	20,0%
Jen Barber	45.600	1	20,0%
Roy Trenneman	84.000	1	43,0%
Total	**481.690**	**6**	**197,0%**

```
Subtotal(9, {1}:{1})
```

After

	Wages and salaries	FTE	Fixed commission %
	50300	HR4000	PYL_940
Rick Deckard	91.340	1	35,0%
Roy Batty	135.050	1	45,0%
Ellen Ripley	70.400	1	34,0%
Charlie Bucket	55.300	1	20,0%
Jen Barber	45.600	1	20,0%
Roy Trenneman	84.000	1	43,0%
Total	**481.690**	**6**	**32,8%**

```
Subtotal(9, {1}:{1})
```

```
Subtotal(1, {1}:{1})
```

Use of placeholder in Row/Column 1 -

+

Give a reference for in-matrix calculations
Position in matrix not impacted from dynamicity
Simplify formula syntax and enhance readability

−

Apparently complicated matrix structure
Solution dependent on header

Useful Excel formulas

ADDRESS(1, 1) → A1

INDIRECT (A1) → Retrieves value held in cell A1

INDIRECT(ADDRESS(1,1)) → Retrieves value held in cell A1

ROW: Retrieves row of referenced cell

ROW(B2) → 2

ROW() → Retrieves the row where the formula is

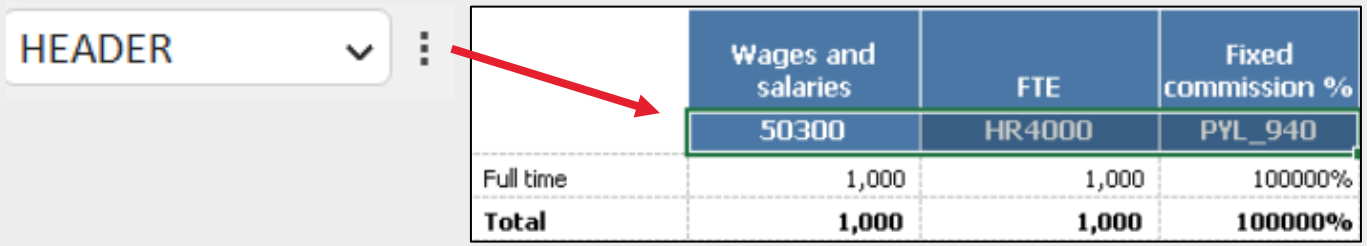
COLUMN: Retrieves column of referenced cell

COLUMN(D4) → 4

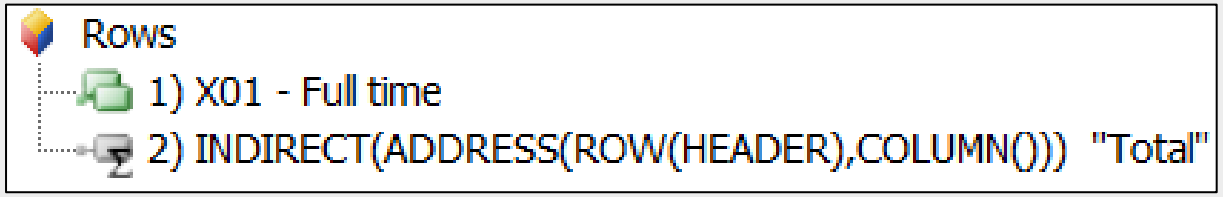
COLUMN() → Retrieves the column where the formula is

Solution B: use of named range

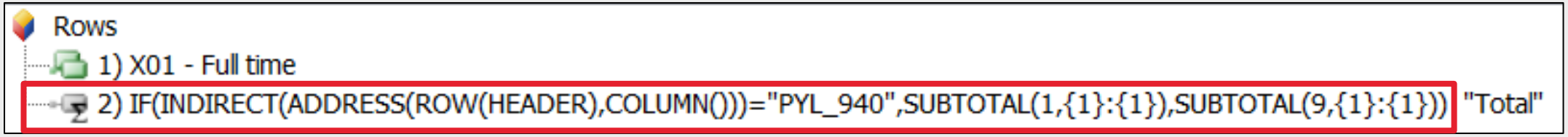
1) Create a Named Range element in the row which contains the header (Account code)



2) Retrieve the header value based on the range row and the current column using Indirect/Address



3) Change the formula according to the value on the header row



Use of named range

More maintainable than Placeholder use

Formulas don’t break if matrix is moved

Vlookup function: Example

Project

Project ID	Project manager	Project plan type
P200	Chuck Billy	Hold
P100	Dave Mustaine	Grow
P400	Steven Wilson	Disinvest
P300	Robert Ellsworth	Disinvest
P600	Rob Halford	Grow
P500	Scott Ian	Hold
P800	John Petrucci	Grow
P700	Steve Harris	Hold

Syntax: **Vlookup**(Text to find, Area, Number of column to retrieve, FALSE)

Example: Vlookup(P100, Project, 3, FALSE) → Grow

Example 2: Searching elements in different tables

VAT

Entity	VAT
A01	17.00%
A00	18.00%
B00	17.00%
A02	20.00%
C00	22.00%
B01	23.00%
C02	16.00%
C01	21.00%
D00	22.00%
C03	20.00%

Selling price

			Average price	VAT	VAT Amount	Selling price
A00	Stark Industries	TV Spare Parts	780		0	780
A00	Stark Industries	Smartphones Spare Parts	300		0	300
B01	Wayne Enterprises	TV Spare Parts	650		0	650
B01	Wayne Enterprises	Smartphones Spare Parts	290		0	290
A02	A.C.M.E. Co.	TV Spare Parts	680		0	680
A02	A.C.M.E. Co.	Smartphones Spare Parts	310		0	310
B00	Monsters Inc.	TV Spare Parts	670		0	670
B00	Monsters Inc.	Smartphones Spare Parts	320		0	320
C00	Gringotts	TV Spare Parts	730		0	730
C00	Gringotts	Smartphones Spare Parts	250		0	250

Example 2: Searching elements in different tables

VAT

Entity	VAT
A01	**17.00%**
A00	**18.00%**
B00	**17.00%**
A02	**20.00%**
C00	**22.00%**
B01	**23.00%**
C02	**16.00%**
C01	**21.00%**
D00	**22.00%**
C03	**20.00%**

Selling price

			Average price	VAT	VAT Amount	Selling price
A00	Stark Industries	TV Spare Parts	780	**18.00%**	140	920
A00	Stark Industries	Smartphones Spare Parts	300	**18.00%**	54	354
B01	Wayne Enterprises	TV Spare Parts	650	**23.00%**	150	800
B01	Wayne Enterprises	Smartphones Spare Parts	290	**23.00%**	67	357
A02	A.C.M.E. Co.	TV Spare Parts	680	**20.00%**	136	816
A02	A.C.M.E. Co.	Smartphones Spare Parts	310	**20.00%**	62	372
B00	Monsters Inc.	TV Spare Parts	670	**17.00%**	114	784
B00	Monsters Inc.	Smartphones Spare Parts	320	**17.00%**	54	374
C00	Gringotts	TV Spare Parts	730	**22.00%**	161	891
C00	Gringotts	Smartphones Spare Parts	250	**22.00%**	55	305

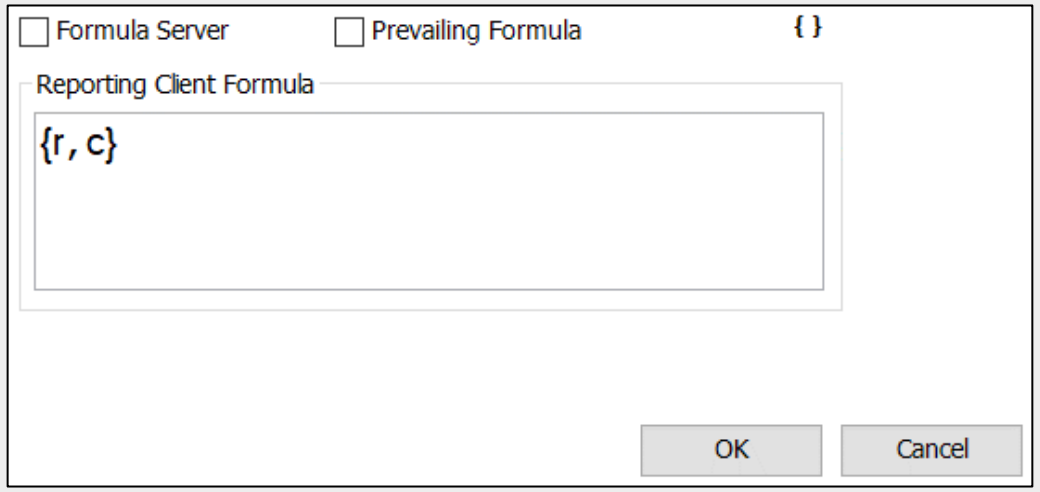
Solution steps

- Define VAT Area
- Use Vlookup formula to retrieve VAT in Selling Price

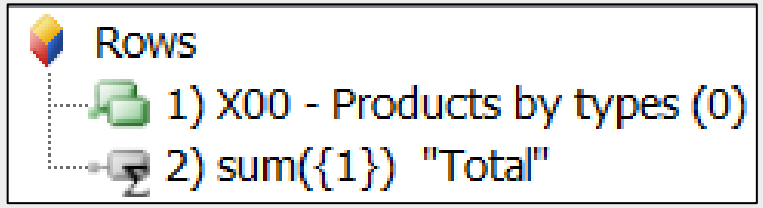
Bi-dimensional reference in matrix formula

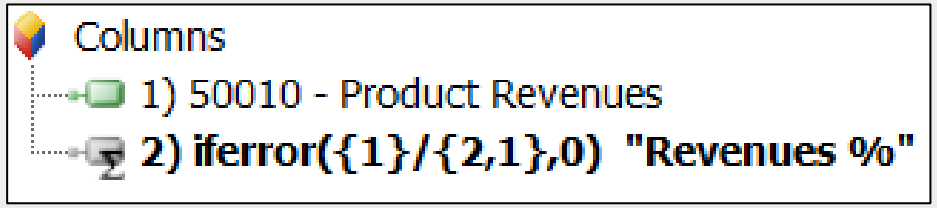
- Allows us to refer to a matrix cell specifying its coordinates (row, column)
- Prevent the use of Excel areas or complex formula as INDIRECT or OFFSET
- Formula is translated in a simple Excel reference

Syntax used in matrix wizard for Reporting client formula on row/column filters : **{r,c}**

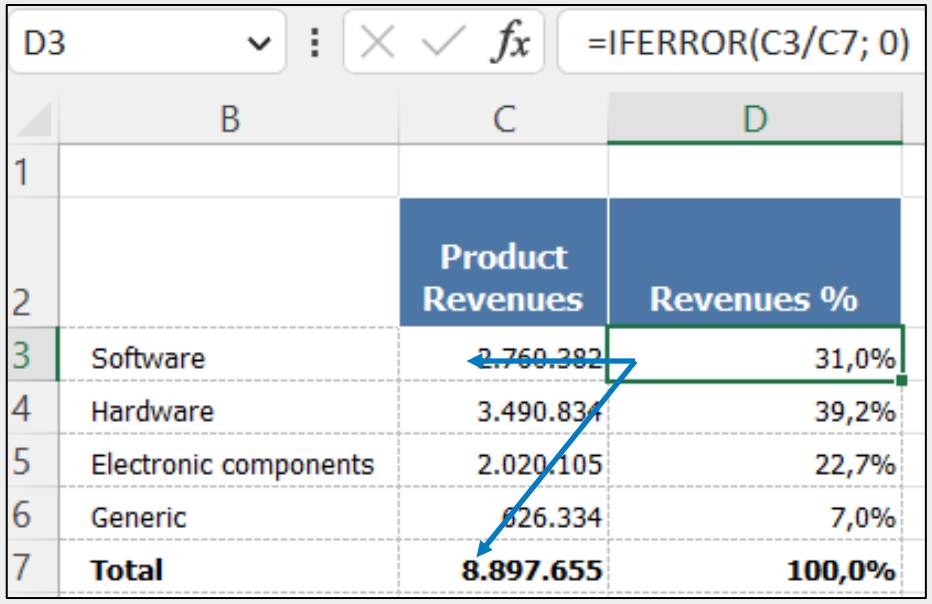


Bi-dimensional reference in matrix formula - Example





Where **{2,1}** represents the total of the product revenues (row 2, column 1)



	B	C	D
1			
2		Product Revenues	Revenues %
3	Software	2.760.382	31,0%
4	Hardware	3.490.834	39,2%
5	Electronic components	2.020.105	22,7%
6	Generic	626.334	7,0%
7	**Total**	**8.897.655**	**100,0%**

Use of In-Form Calculations

Use

- Calculations related to one area of interest
- If other calculation tools cannot achieve the desired result
- Immediate Feedback Required

Don't Use

- Calculation depends on values from other forms or steps
- Calculation must run on large volume of data or elements
- Submission depends on the calculation being run

What can change in a form?



I add a new matrix



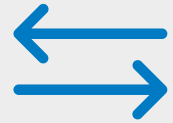
I move a matrix



I add row/columns between a matrix

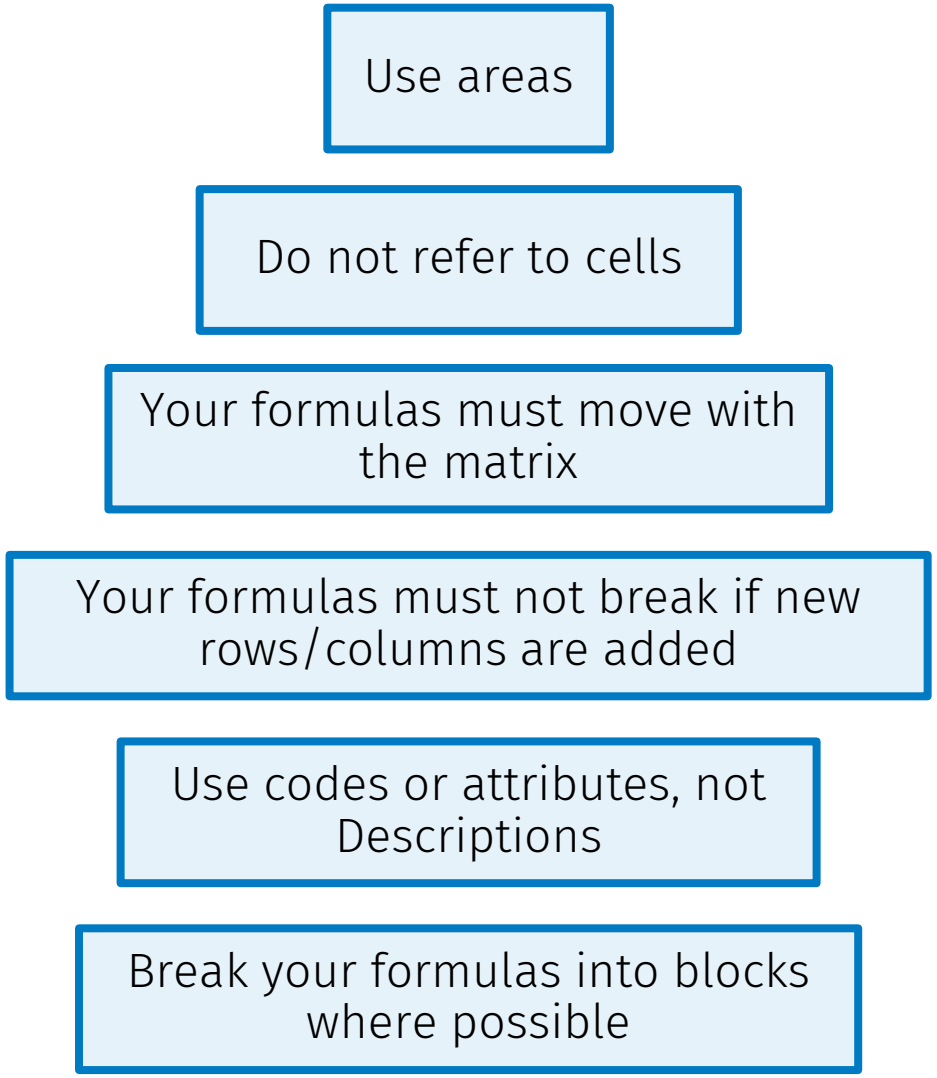


My Matrix expands based on new dynamic elements



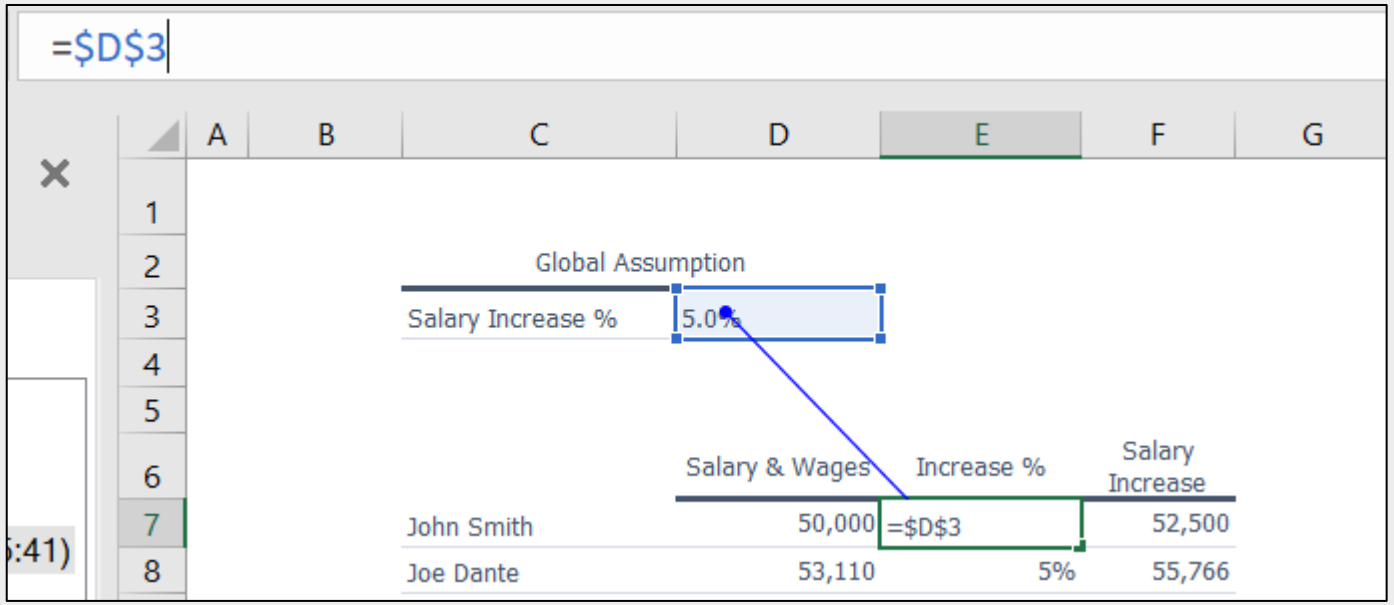
I change the description/attribute of a dimension element

Best Practices/Recommendations from the field



Problem 1

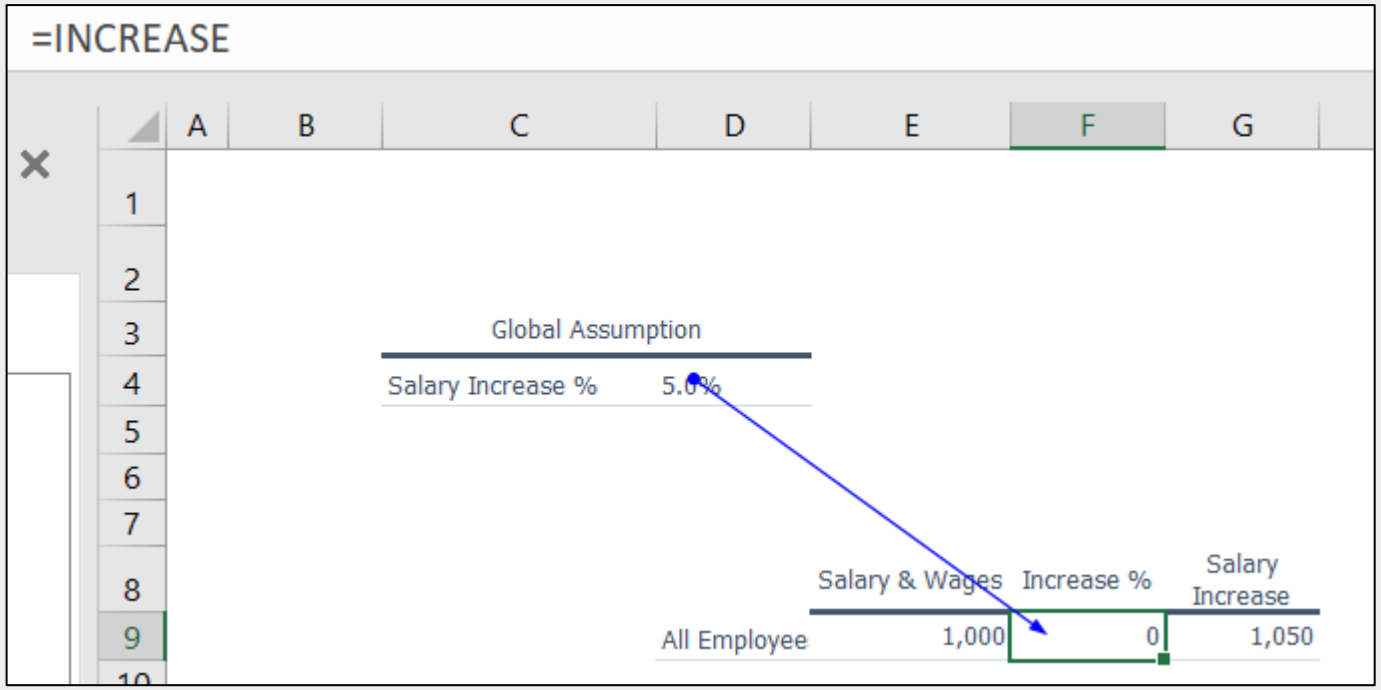
Referring directly to a cell address...



If you shift the position of the matrix or enter a new row/column, the formula will no longer work

Solution

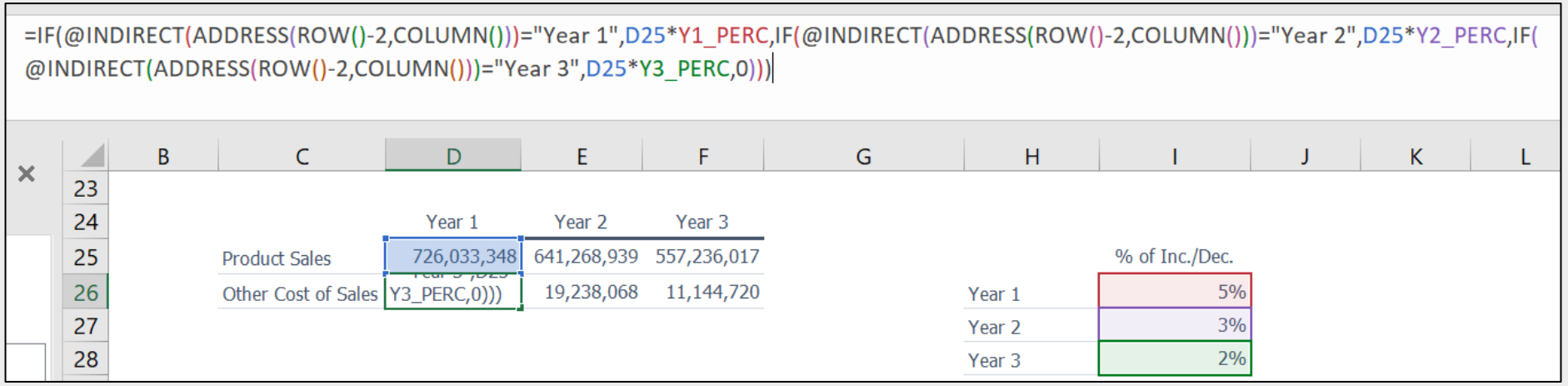
Referring directly to a cell address → Use Ranges!



The formula will now be independent of the position of the value being referred

Problem 2

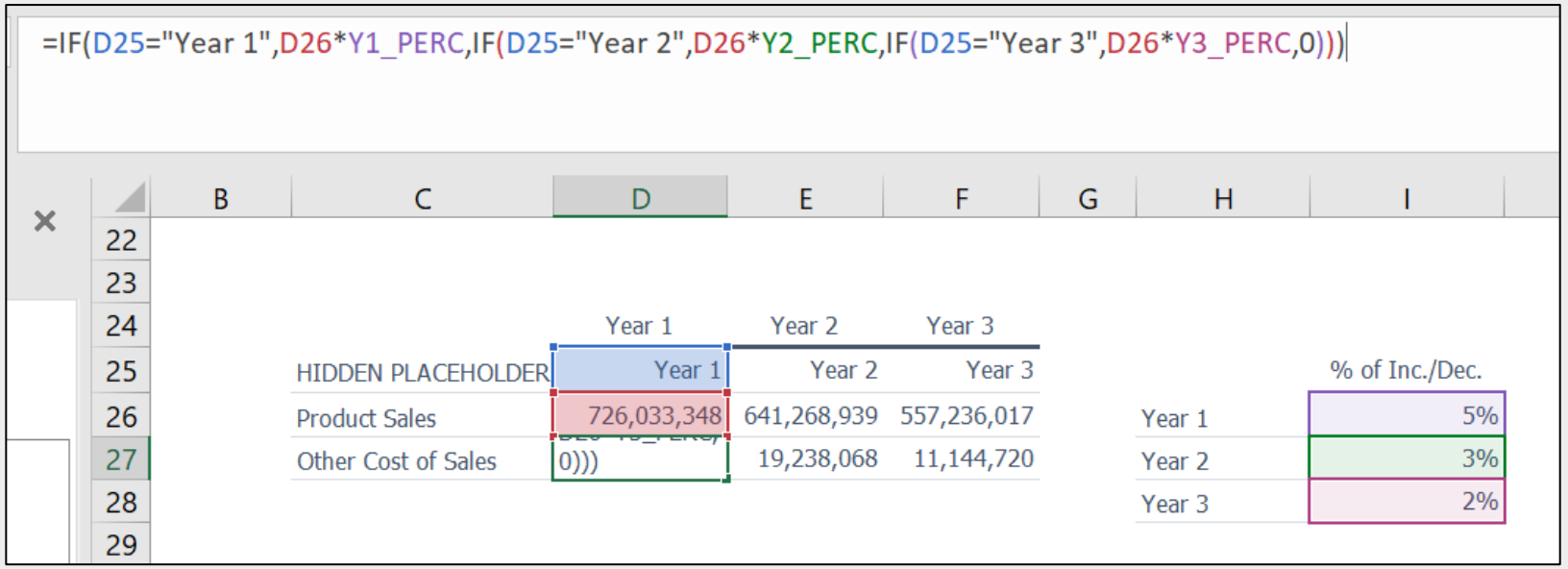
Repeating the same statement when referring to a column



The Indirect assignment in this case is repeated multiple times making the code difficult to read...

Solution

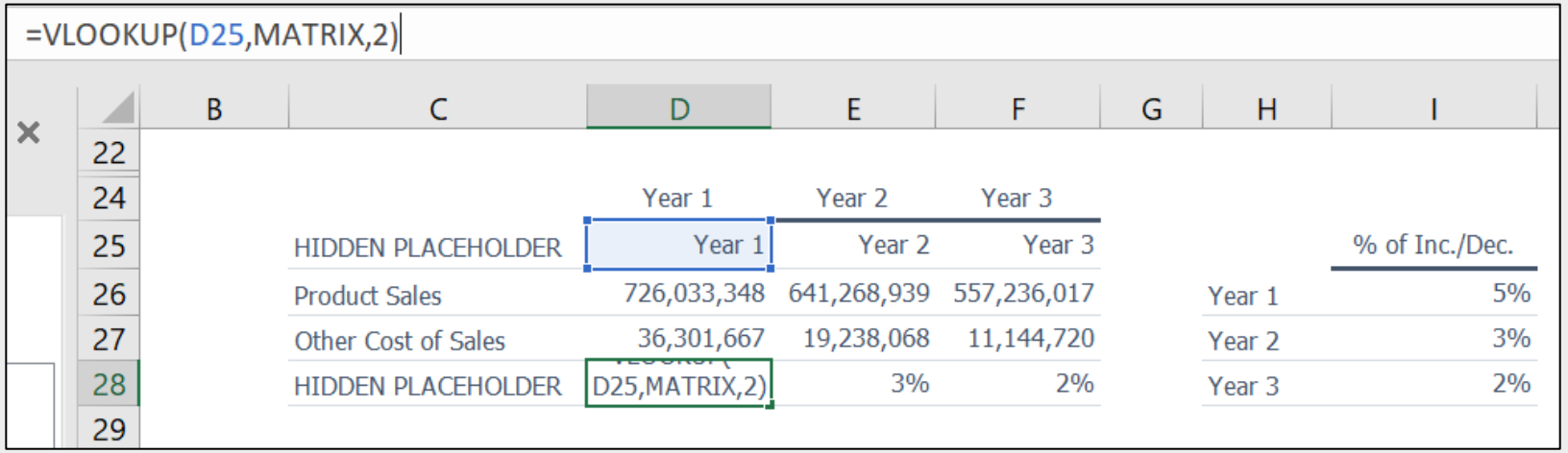
Repeating the same statement when referring to a column → Use Placeholders



By moving the indirect expression to a hidden placeholder our formula is easier to read and debug

Solution

Repeating the same statement when referring to a column → Use Lookup Formulas



Where possible use Lookup formulas instead of nested IF statements to reduce complexity

Business case

Project

Project ID	Project manager	Project plan type
P200	Chuck Billy	Hold
P100	Dave Mustaine	Grow
P400	Steven Wilson	Disinvest
P300	Robert Ellsworth	Disinvest
P600	Rob Halford	Grow
P500	Scott Ian	Hold
P800	John Petrucci	Grow
P700	Steve Harris	Hold

Capital costs

		2048 Budget	2049 Budget	2049 Budget	2049 Budget
		Capital costs	Increase %	Capital costs	Project contribution %
		50800	50328	50800	
P100	Office upgrades	3.693			
P200	CPM Tool evaluation	14.840			
P300	Market Buyers' analysis	30.380			
P400	Laptop upgrades	9.540			
P500	New servers	38.760			
P600	Network cabling	4.500			
P700	Wall Street campaign	15.007			
P800	CRM Upgrade	7.890			
Total		**124.610**			

Global Assumptions

Action	Percentage
Hold	0,5%
Grow	5,0%
Disinvest	-1,0%

Business case: Solution steps

1. Retrieve Project plan type from Capital costs Project code

Project

Project ID	Project manager	Project plan type
P200	Chuck Billy	Hold
P100	Dave Mustaine	Grow
P400	Steven Wilson	Disinvest
P300	Robert Ellsworth	Disinvest
P600	Rob Halford	Grow
P500	Scott Ian	Hold
P800	John Petrucci	Grow
P700	Steve Harris	Hold

Capital costs

		2048 Budget	2049 Budget	2049 Budget	2049 Budget
		Capital costs	Increase %	Capital costs	Project contribution %
		50800	50328	50800	
P100	Office upgrades	3.693			
P200	CPM Tool evaluation	14.840			
P300	Market Buyers' analysis	30.380			
P400	Laptop upgrades	9.540			
P500	New servers	38.760			
P600	Network cabling	4.500			
P700	Wall Street campaign	15.007			
P800	CRM Upgrade	7.890			
Total		**124.610**			

Global Assumptions

Action	Percentage
Hold	0,5%
Grow	5,0%
Disinvest	-1,0%

Business case: Solution steps

1. Retrieve Project plan type from Capital costs Project code
2. Retrieve Percentage from Global assumption

Project

Project ID	Project manager	Project plan type
P200	Chuck Billy	Hold
P100	Dave Mustaine	Grow
P400	Steven Wilson	Disinvest
P300	Robert Ellsworth	Disinvest
P600	Rob Halford	Grow
P500	Scott Ian	Hold
P800	John Petrucci	Grow
P700	Steve Harris	Hold

Global Assumptions

Action	Percentage
Hold	0,5%
Grow	5,0%
Disinvest	-1,0%

Capital costs

		2048 Budget	2049 Budget	2049 Budget	2049 Budget
		Capital costs	Increase %	Capital costs	Project contribution %
		50800	50328	50800	
P100	Office upgrades	3.693			
P200	CPM Tool evaluation	14.840			
P300	Market Buyers' analysis	30.380			
P400	Laptop upgrades	9.540			
P500	New servers	38.760			
P600	Network cabling	4.500			
P700	Wall Street campaign	15.007			
P800	CRM Upgrade	7.890			
Total		**124.610**			

Business case: Solution steps

1. Retrieve Project plan type from Capital costs Project code
2. Retrieve Percentage from Global assumption
3. Apply the right percentage in Capital costs

Project

Project ID	Project manager	Project plan type
P200	Chuck Billy	Hold
P100	Dave Mustaine	Grow
P400	Steven Wilson	Disinvest
P300	Robert Ellsworth	Disinvest
P600	Rob Halford	Grow
P500	Scott Ian	Hold
P800	John Petrucci	Grow
P700	Steve Harris	Hold

Global Assumptions

Action	Percentage
Hold	0,5%
Grow	5,0%
Disinvest	-1,0%

Capital costs

		2048 Budget	2049 Budget	2049 Budget	2049 Budget
		Capital costs	Increase %	Capital costs	Project contribution %
		50800	50328	50800	
P100	Office upgrades	3.693	5,0%	3.878	3,09%
P200	CPM Tool evaluation	14.840	0,5%	14.914	11,90%
P300	Market Buyers' analysis	30.380	-1,0%	30.076	23,99%
P400	Laptop upgrades	9.540	-1,0%	9.445	7,53%
P500	New servers	38.760	0,5%	38.954	31,07%
P600	Network cabling	4.500	5,0%	4.725	3,77%
P700	Wall Street campaign	15.007	0,5%	15.082	12,03%
P800	CRM Upgrade	7.890	5,0%	8.285	6,61%
Total		**124.610**	**0,6%**	**125.358**	**100,00%**

Business case: Solution steps

1. Retrieve Project plan type from Capital costs Project code
2. Retrieve Percentage from Global assumption
3. Apply the right percentage in Capital costs
4. Configure the formulas correctly to have the right figures

Project

Project ID	Project manager	Project plan type
P200	Chuck Billy	Hold
P100	Dave Mustaine	Grow
P400	Steven Wilson	Disinvest
P300	Robert Ellsworth	Disinvest
P600	Rob Halford	Grow
P500	Scott Ian	Hold
P800	John Petrucci	Grow
P700	Steve Harris	Hold

Global Assumptions

Action	Percentage
Hold	0,5%
Grow	5,0%
Disinvest	-1,0%

Capital costs

		2048 Budget	2049 Budget	2049 Budget	2049 Budget
		Capital costs	Increase %	Capital costs	Project contribution %
		50800	50328	50800	
P100	Office upgrades	3.693	5,0%	3.878	3,09%
P200	CPM Tool evaluation	14.840	0,5%	14.914	11,90%
P300	Market Buyers' analysis	30.380	-1,0%	30.076	23,99%
P400	Laptop upgrades	9.540	-1,0%	9.445	7,53%
P500	New servers	38.760	0,5%	38.954	31,07%
P600	Network cabling	4.500	5,0%	4.725	3,77%
P700	Wall Street campaign	15.007	0,5%	15.082	12,03%
P800	CRM Upgrade	7.890	5,0%	8.285	6,61%
Total		**124.610**	**0,6%**	**125.358**	**100,00%**

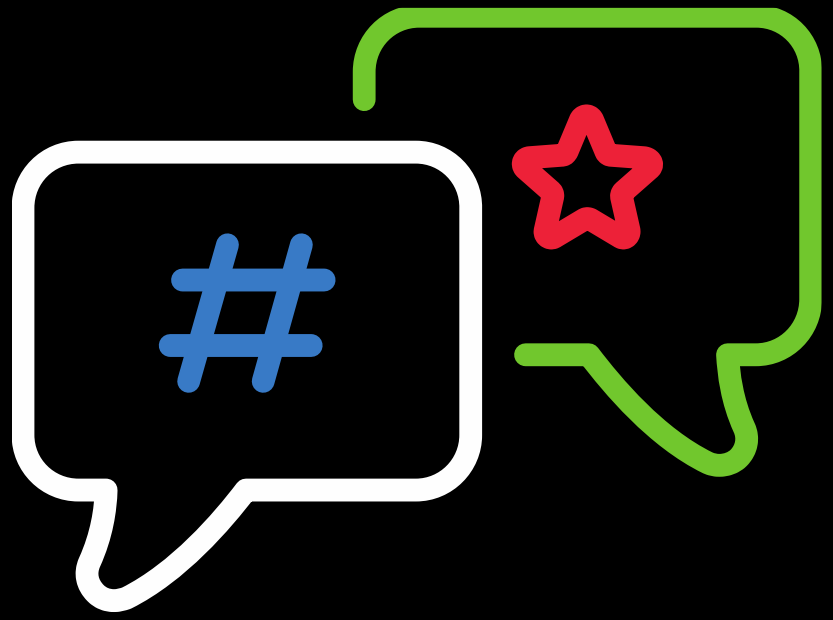
Now it’s your turn

Exercise # 9

Complex data entry forms



Let’s talk about: Validation



Validation

- Prevents us from saving wrong values in a data entry form
- Validation on close or on save
- Definition On Excel cell or in matrix wizard
- A tolerance threshold of validation can be set

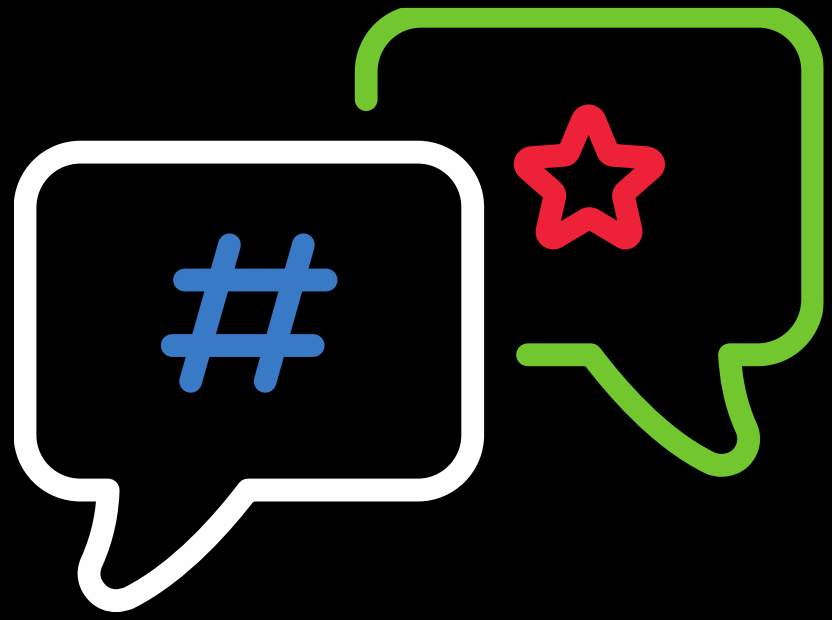
Now it’s your turn

Exercise # 10

Validation



Let’s talk about: Data processing & Data entry forms



Data processing run from data entry form

When

- When data collected in the form are the input of a data calculation (Asset depreciation, Allocation, DTP…)
- When Form must display the outcome of a Data processing

Running

- On demand by hyperlink or by a button
- Automatic run on saving data

Data processing run from data entry form: Parameters

Parameters

- **Excel cell:** value retrieved from an Excel cell
- **Automatic:** inherited from the environment
- **Select:** selection from a list of possible values
- **Insert:** type the code on a textbox
- **Cell field:** Definition from cell field editor

Other useful options:

- Force refresh
- Automatic run on saving data
- At runtime show a link for the run

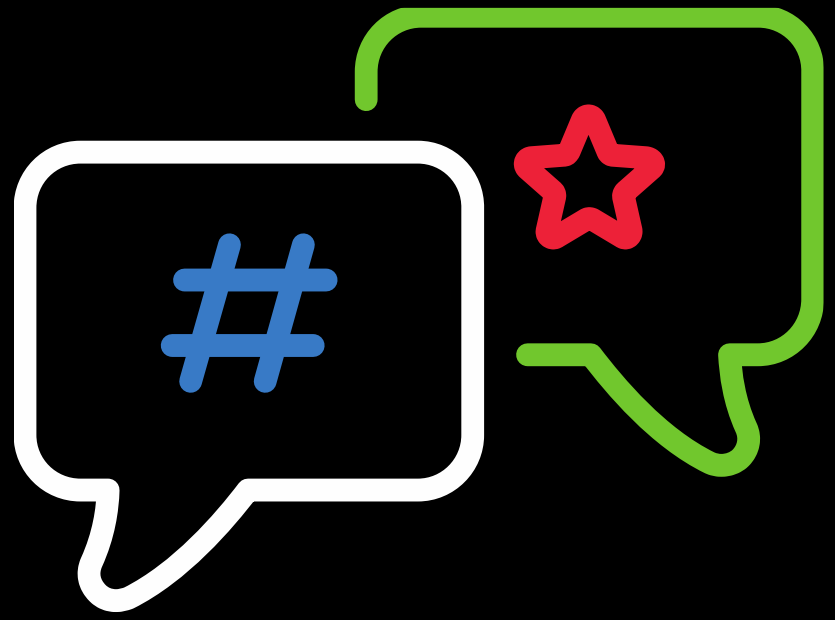
Now it’s your turn

Exercise # 11

Data processing & Data entry forms



Let’s talk about: Drill through – Cell fields



Data analysis on FW: Drill down vs. Drill-through

Drill down

More details inside CCH Tagetik data model (FW)

Drill–through

More details outside CCH Tagetik data model

- Data from Staging table (ETL)
- Data from Datasets (AIH)

Drill-through query

Defined as a Query

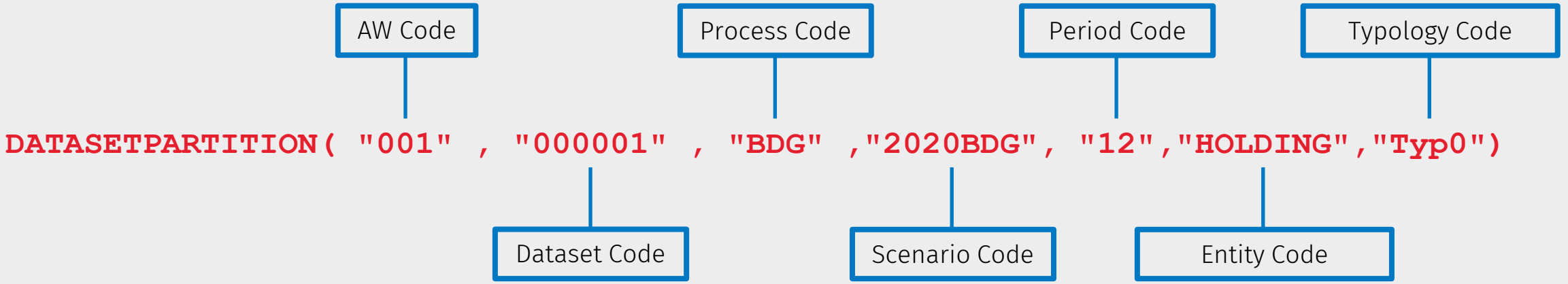
Run on demand and not along with form

Parameters retrieved from dimensions of selection

Physical Name of a Dataset Partition

Cell field - DATASETPARTITION ():

in drill-through queries, DTP, Datasource mapping, SQL clause retrieves the name of dataset partition based on its names and context.



PROCESS($Scenario.code,$Period.code).code → Useful to retrieve Process code

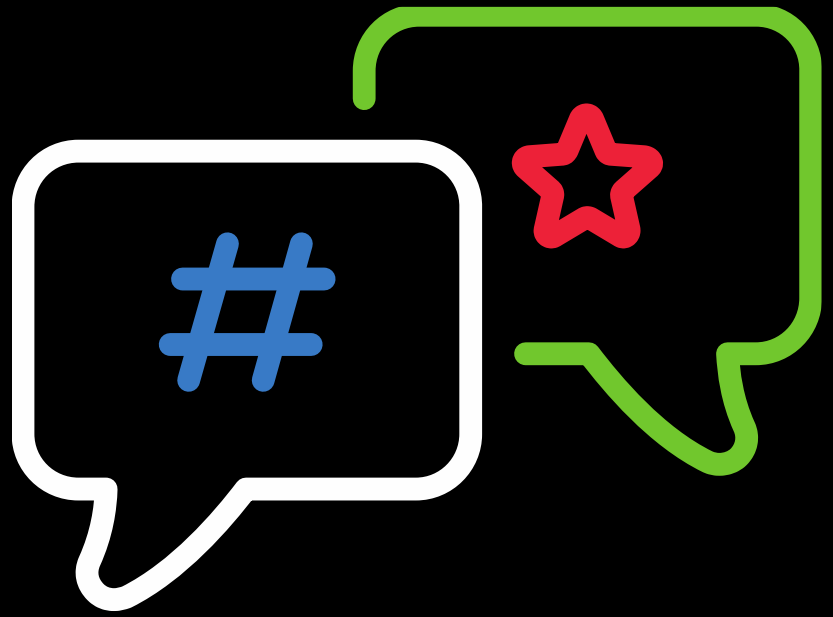
Now it’s your turn

Exercise # 12

Drill-through query



Let’s talk about: Settings - Form



Settings - Form 1/2

General settings inherited from all the forms.

Configuration affects:

Default

- Segment
  - Parameter selection mode
  - Conversion/Filter based on conversion currency

Display

- Form tree
  - Dimension tree

Data entry

- Make the form editable for conversion on the fly

Settings - Form 2/2

Comments

- Automatic application of comments in DE and navigation forms (impact on performance)

Performance

Limit the rows:

- generated by query
- of elements displayed as prompt filters

Run form options (Form Optimization)

- Default settings inherited in each form. Override can be applied in form options.

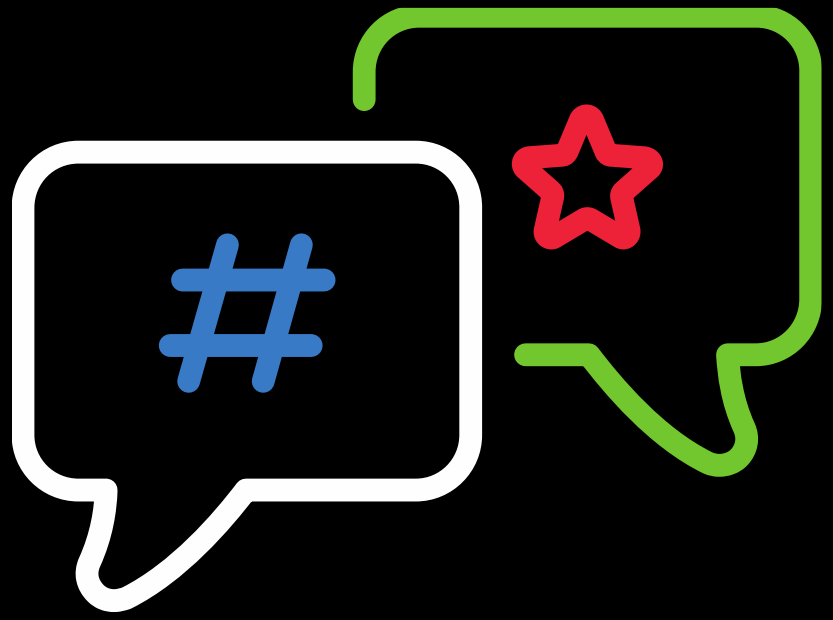
Now it’s your turn

Exercise # 13

Web configuration



Let’s talk about: Override function



Override function

Typical use

- **Force** the editability settings or the content of a cell with a formula or a value (Header)
- **Manage exception for single cells** not entire row or column of a matrix
- **Used in static matrix** and not dynamic

Practical info

- Managed on Open items
- Overridden cells displayed in red
- Remove by right click

Recommendations

- **Use only when strictly necessary**
- Massive use slows down the form run

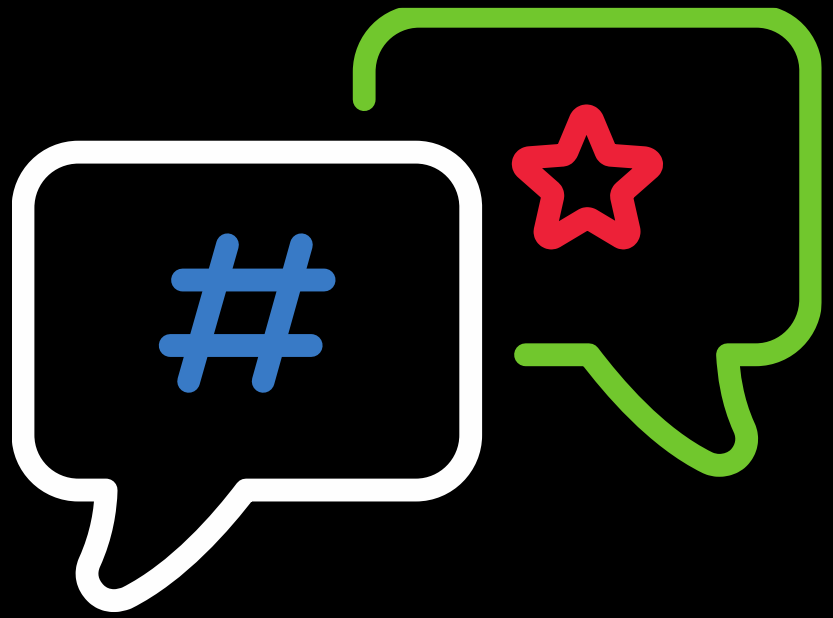
Now it’s your turn

Exercise # 14

Override function



Let’s talk about: FST OnScenarioPeriod utility



FST OnScenarioPeriod utility

Problem

To calculate KPIs by referring to Scenario/Period different from the Scenario/Period given in the form that holds the FST.

Solution

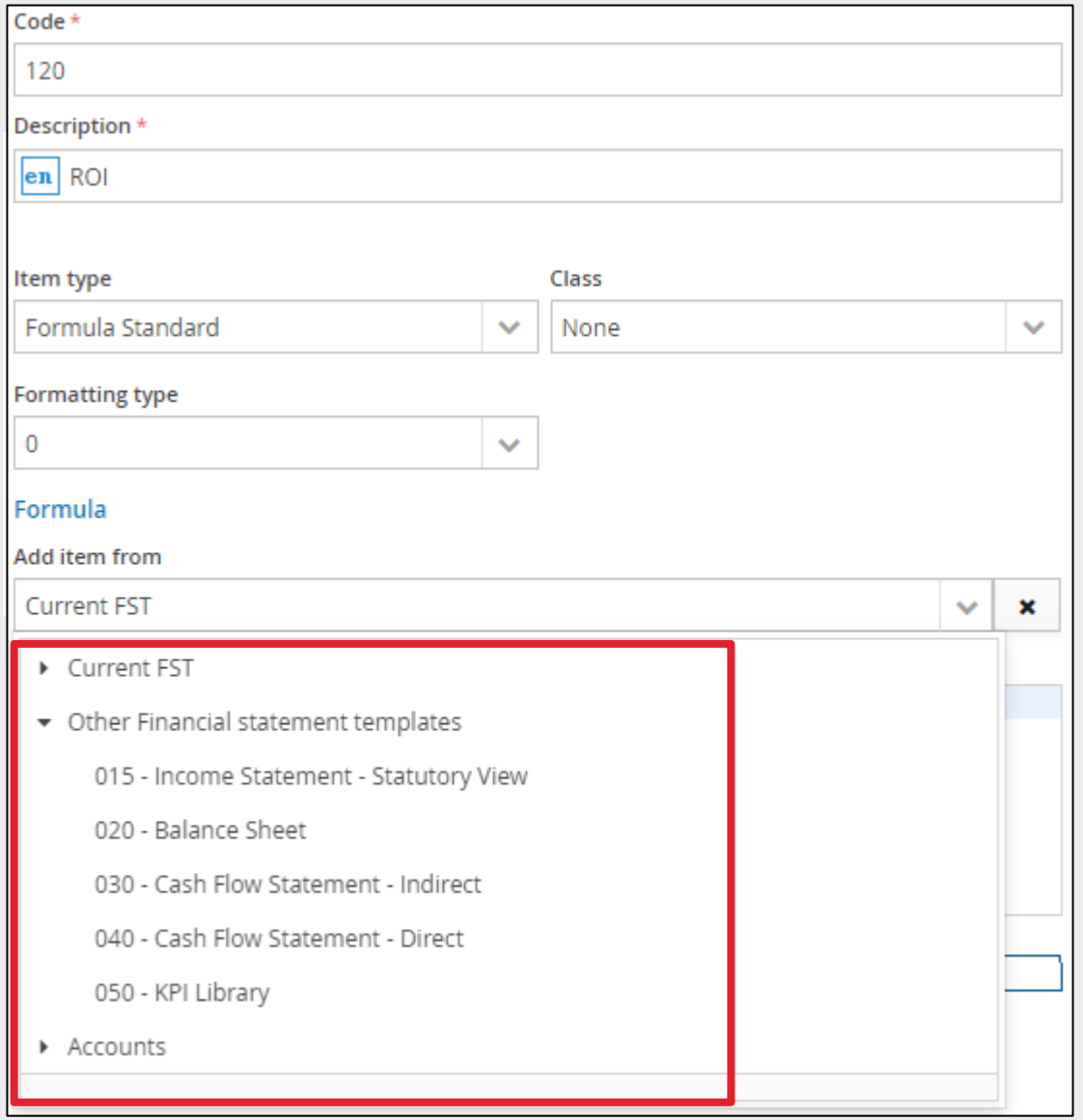
Create an FST Formula using OnScenarioPeriod, applicable to FST item and account.

Example

Syntax	Value retrieved
[SLS|010].onScenarioPeriod(-1)	FST Item 010 related to previous period
[SLS|010].onScenarioPeriod(“2057ACT”,”12”)	FST Item related to 2057ACT 12

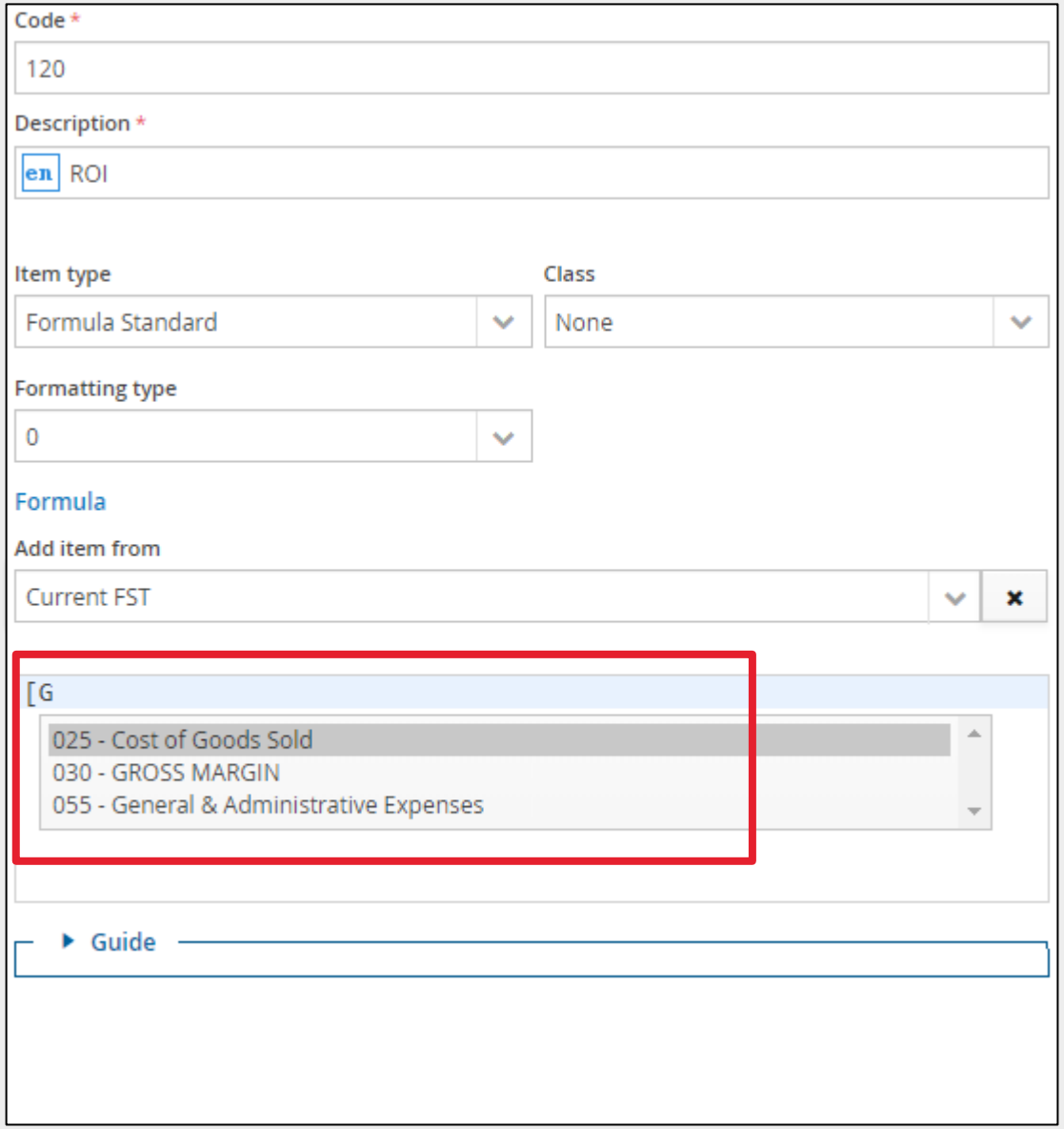
Formula assistant

- For item type Formula Standard
- Selection from
  - Current FST
  - Other FSTs (possibility to select one)
  - Accounts
- Using square bracket and typing code or description



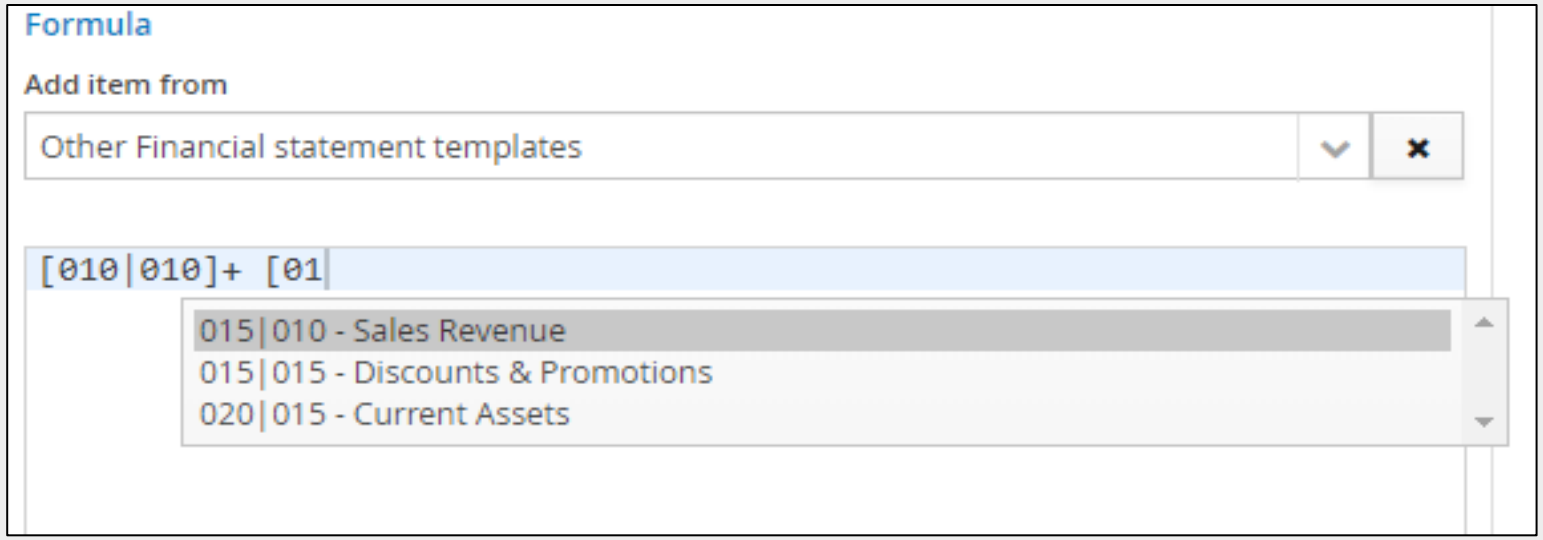
Formula assistant

- For item type Formula Standard
- Selection from
  - Current FST
  - Other FSTs (possibility to select one)
  - Accounts
- Using square bracket and typing code or description



Syntax

- [FST code | FST item code]
- [Account code]
- You can copy and paste



Now it’s your turn

Exercise # 15

FST
onScenarioPeriod
utility



Advanced Reporting extra courses

Report Bursting course is **mandatory** to be Platform Specialist certified.

PowerBI Connector course is **optional**.



Takeaway

Join CCH Tagetik Insiders Hub!

Get access to exclusive CCH Tagetik best practices and tips, sneak peaks at product updates

Share the love by participating in Testimonials, Case studies, Surveys, Reviews, Referrals

Leverage your social capital by connecting with other finance and operations professionals

Earn points up the leader board and redeem rewards

Here's the way the hub works

1 Log into the CCH Tagetik Community and click on the Insiders tab

2 Poke around and complete challenges earning points to redeem rewards

3 Get recognized for all the support you give CCH Tagetik

It's that easy!

Join CCH Tagetik Insiders now!

What other learning assets are available?

User Manual

CCH Tagetik Community

Knowledge Base

What's next?

Visit the Academy webpage

Next training and/or Assessment

Mail to info.academy@wolterskluwer.com

Thank you



Wolters Kluwer

CCH® Tagetik | ACADEMY